

Projections of Solids:

A solid is a 3-D object having length, breadth and thickness and bounded by surfaces which may be either plane or curved, or combination of the two.

Solids are classified under two main headings

- Polyhedron
- Solids of revolution

A regular polyhedron is solid bounded only by plane surfaces (faces). Its faces are formed by regular polygons of same size and all dihedral angles are equal to one another. When faces of a polyhedron are not formed by equal identical faces, they may be classified into prisms and pyramids.

Five regular polyhedron are shown in figure 1

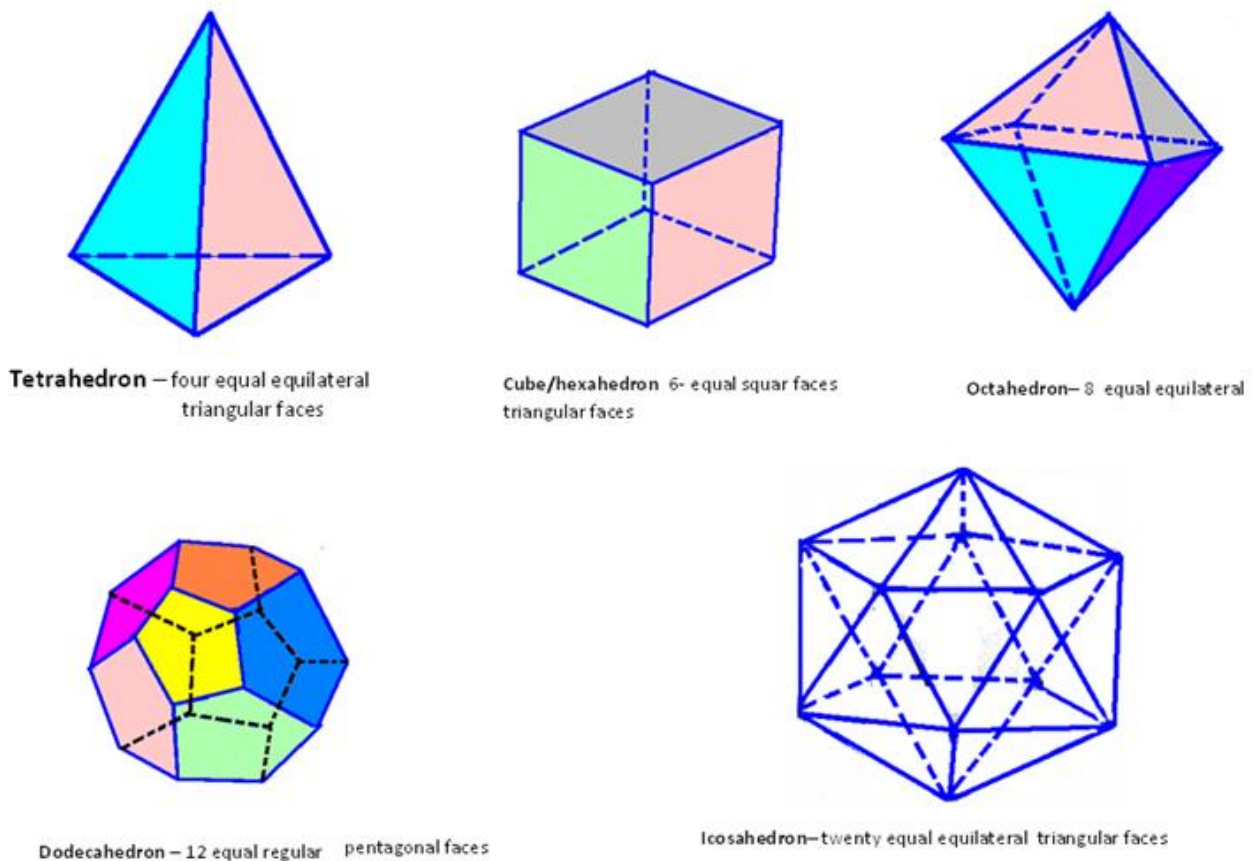


Figure 1: Five regular polyhedra

Prism

Prisms are polyhedron formed by two equal parallel regular polygon, end faces connected by side faces which are either rectangles or parallelograms. Different types of prisms are shown in figure 2

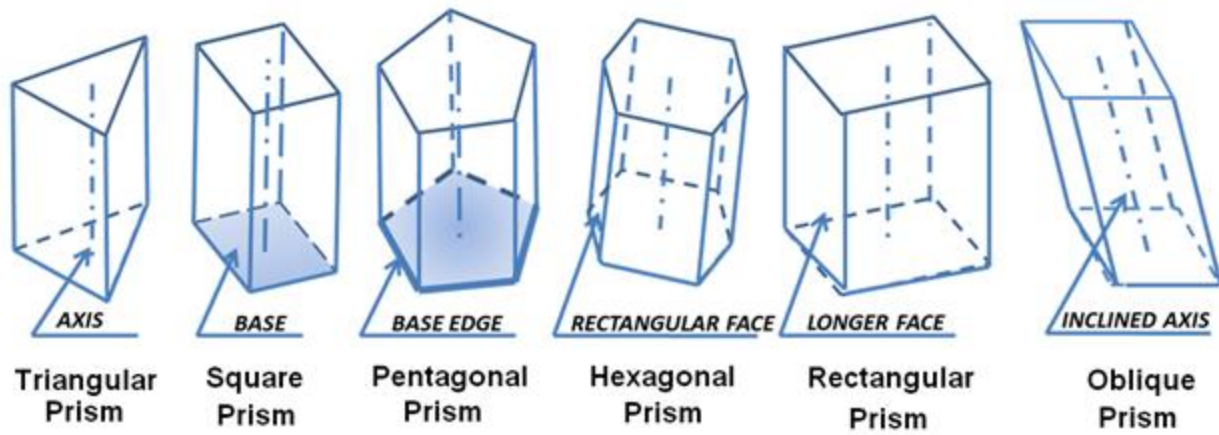


Figure 2. Various types of prisms generally encountered in engineering applications

Some definitions regarding prisms

Base and lateral faces. When the prism is placed vertically on one of its end faces, the end face on which the prism rests is called the base. The vertical side faces are the lateral faces, as shown in Figure 3.

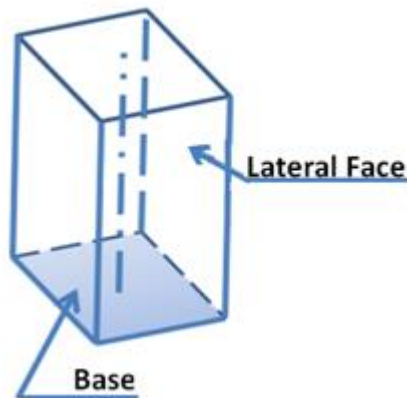


Figure 3. Shows the base and lateral face of a prism.

Base edge/Shorter edge: These are the sides of the end faces, as shown in figure 4.

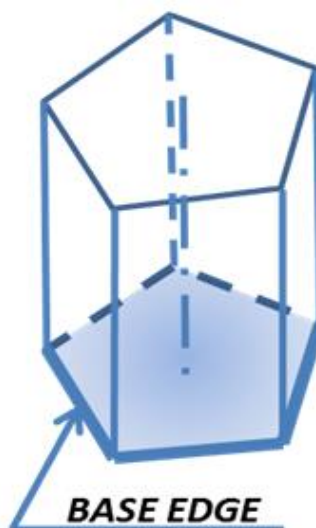


Figure 4. showing the base edge or shorter edge of a pentagonal prism.

Axis – it is the imaginary line connecting the end faces is called axis and is shown in figure 5.

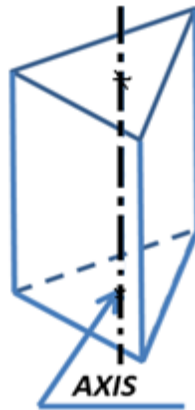


Figure 5 showing the Axis of a triangular prism.

Longer edge/lateral edges: These are the edges connecting the respective corners of the two end faces. The longer edge of a square prism is illustrated in figure 6.

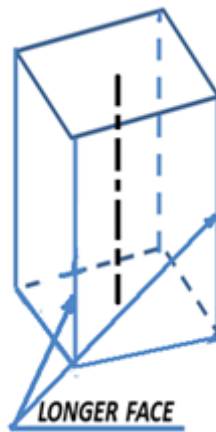


Figure 6. Illustrating the longer edge of a square prism.

Right prism – A prism whose axis is perpendicular to its end face is called as a right prism. Prisms are named according to the shape of their end faces, i.e, if end faces are triangular, prism is called a triangular prism.

Oblique prism: It is the prism in which the axis is inclined to its base.

Pyramids-

Pyramid is a polyhedron formed by a plane surface as its base and a number of triangles as its side faces, all meeting at a point, called vertex or apex.

Axis – the imaginary line connecting the apex and the center of the base.

Inclined/slant faces – inclined triangular side faces.

Inclined/slant/longer edges – the edges which connect the apex and the base corners.

Right pyramid – when the axis of the pyramid is perpendicular to its base.

Oblique pyramid – when the axis of the pyramid is inclined to its base.

Solids of revolution

When some of the plane figures are revolved about one of their sides – solids of revolution is generated some of the solids of revolution are:

1. **Cylinder:** when a rectangle is revolved about one of its sides, the other parallel side generates a cylinder.
2. **Cone:** when a right triangle is revolved about one of its sides, the hypotenuse of the right triangle generates a cone.
3. **Oblique cylinder:** when a parallelogram is revolved about one of its sides, the other parallel side generates a cylinder.
4. **Sphere:** when a semi-circle is revolved about one of its diameter, a sphere is generated..
5. **Truncated and frustums of solids** – when prisms, pyramids, cylinders are cut by cutting planes, the lower portion of the solids (without their top portions) are called, either truncated or frustum of these solids. Some examples are shown in figure 7.

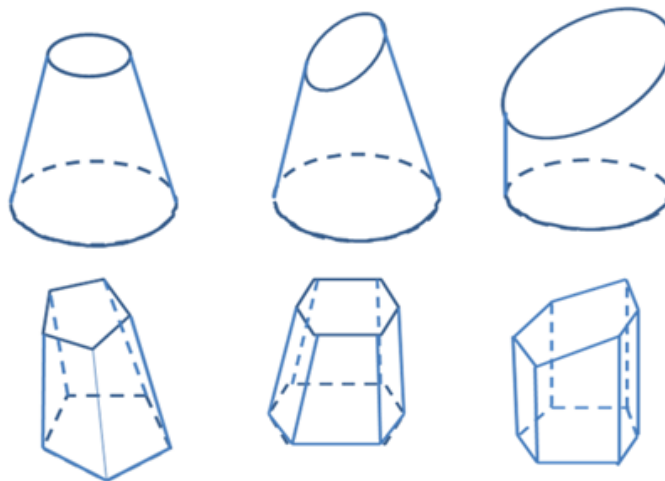


Figure 7. Illustrates some examples of truncated / frustum of solids.

Visibility

When drawing the orthographic views of an object, it will be required to show some of the hidden details as invisible. To distinguish the invisible portions from the visible ones, the invisible edges of the object are shown on the orthographic views by dashed lines. However, in practice, these lines of dashes conveniently and colloquially, but wrongly called as dotted lines. To identify the invisible portions of the object, a careful imaginative thinking is essential.

Rules of visibility

When viewing an object, the outline of the object is visible. Hence the outlines of all the views are shown by full lines. All the visible edges will be shown as solid lines as shown in figure 8. Figure shows the frustum of a pentagonal pyramid.

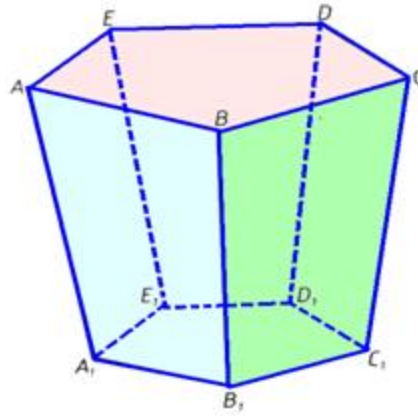


Figure 8. Front view of the object. The visible edges are shown as solid lines and the hidden edges are shown as dashed lines.

Figure 9 shows the projections of the object. In the top view, the highest portions of the object are visible. The top face ABCDE is at the top and is completely visible in the top view. In the top view, edges ab, bc, cd, de and ea are shown as full lines. The bottom pentagonal faces A1B1C1D1E1 is smaller than the top face, hence invisible. The slant edges AA1, BB1, CC1, DD1 and EE1 are invisible in the top view, hence they are shown as lines of dashes. The line connecting a visible point and an invisible point is shown as an invisible line of dashes unless they are out lines.

In the front view, the front faces of the object are shown as visible. The faces ABB1A1 and BCC1B1 are the front faces. Hence in the front view, the corners a, b, c and a1, b1, c1 are visible to the observer. Hence in the front view, the lines a'a'1, b'b'1 and c'c'1 are shown as full lines. However the corners d, e, d1 and e1 are invisible in the front view. The lines, e'e'1, d'd'1 are invisible, hence shown as dashed lines. The top rear edges a'e', e'd' and d'c' coincide with the top front visible edges a'b' and b'c'.

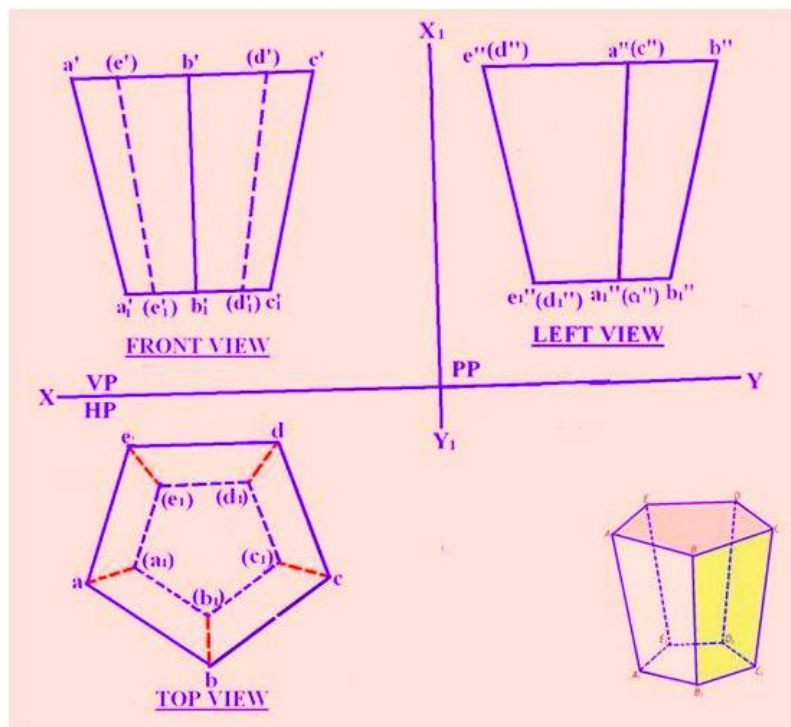


Figure 9. Projections of the frustrum of a pentagonal pyramid.

In the side view - the face lying on that side are visible. As seen in the left side view, the corners e, a, b and e1, a1, b1 lie on left side and are visible in the left view. Hence the lines, e''e1'', a''a1'' and b''b1'' are shown as full lines. The edges d''d1'', c''c1'' coincide with the visible edges e''e1'' and a''a1'' respectively.

Projections of solids placed in different positions

The solids may be placed on HP in various positions

1. The way the axis of the solid is held with respect to HP or VP or both -

- Perpendicular to HP or VP
- Parallel to either HP or VP and inclined to the other
- Inclined to both HP and VP
- The portion of the solid on which it lies on HP, except when it is freely suspended position. It can lie on HP on its base edge or a corner, or a lateral face, or apex.

2. Axis of the solid perpendicular to HP

A solid when placed on HP with its axis perpendicular to it, then it will have its base on HP. This is the simplest position in which a solid can be placed. When the solid is placed with the base on HP position, in the top view, the base will be projected in its true shape. Hence, when the base of the solid is on HP, the top view is drawn first and then the front view and the side views are projected from it. Figure 10 shows a cylinder with its axis perpendicular to HP. There is only one position in which a cylinder or a cone may be placed with its base on HP.

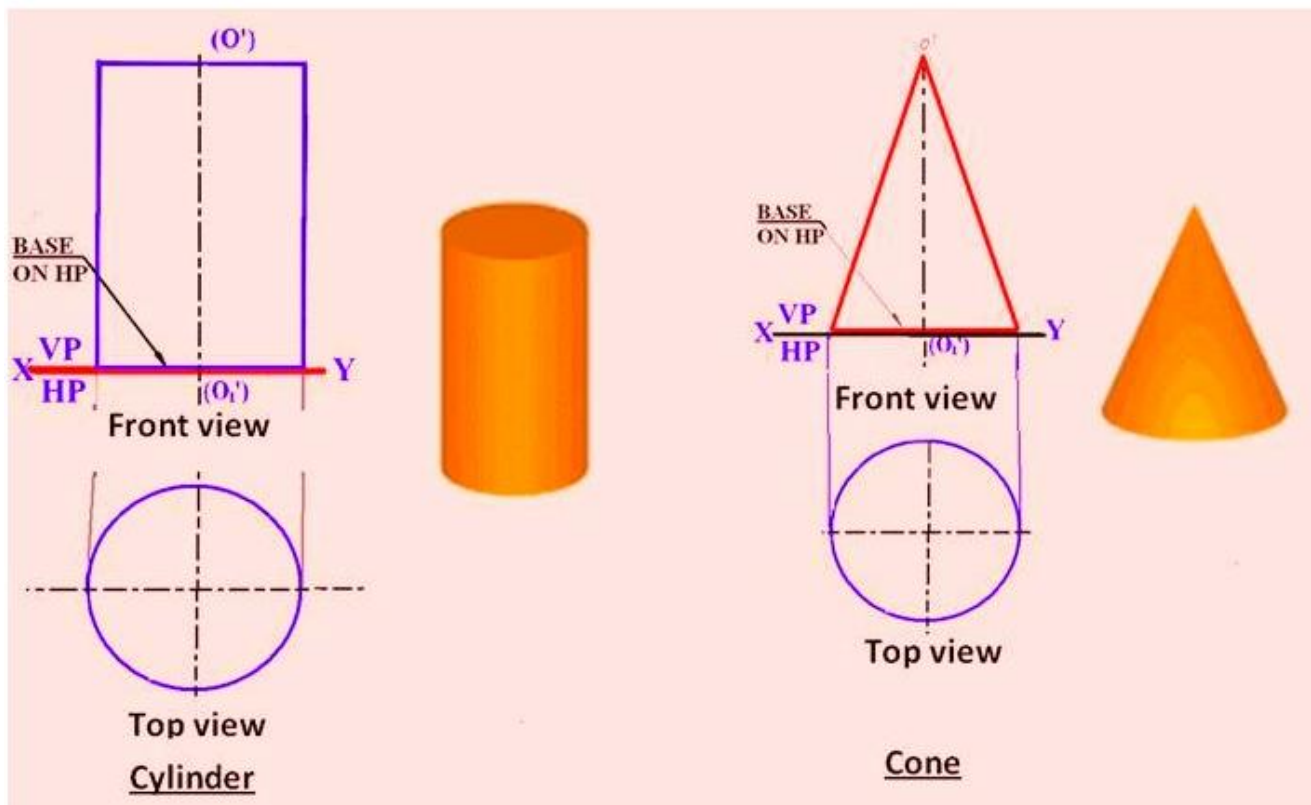


Figure 10. Front view and top view of a cylinder and cone

For prisms, there are 4 positions it may be placed with its base on HP. These positions are illustrated in figure 11.

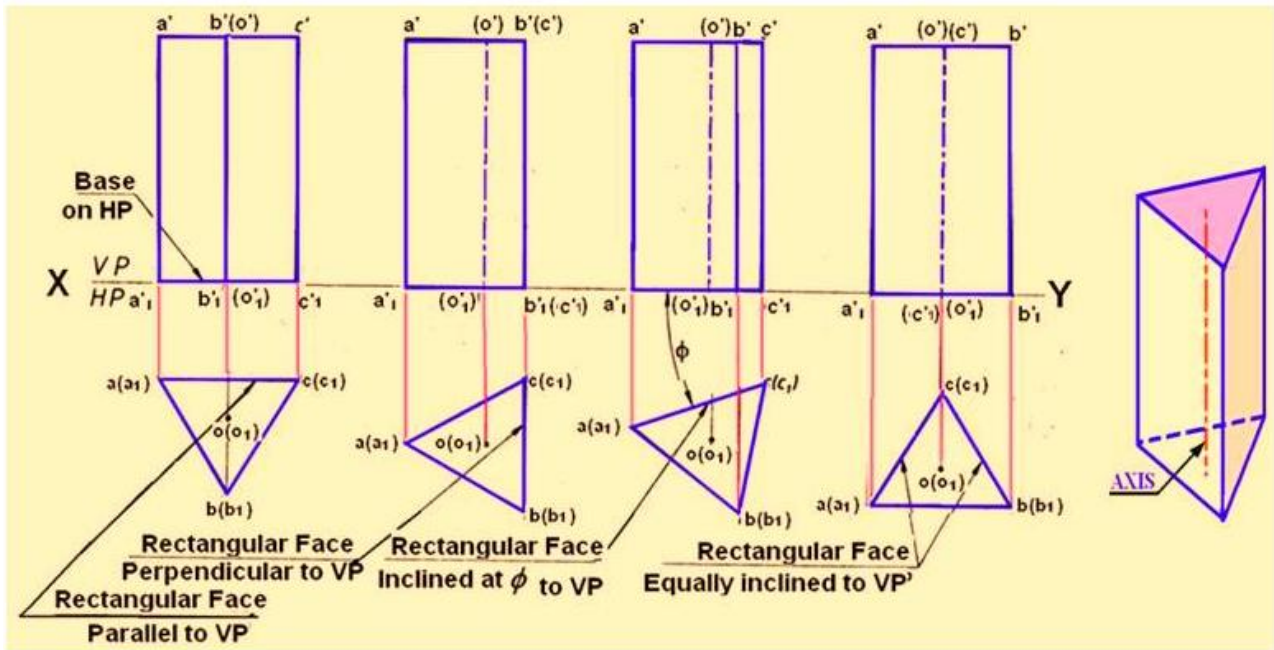


Figure 11. Projections of a triangular prism resting on its base on HP with different positions.

There are 4 positions in which pyramids may be placed with its base on HP. These positions are shown in figure 12.

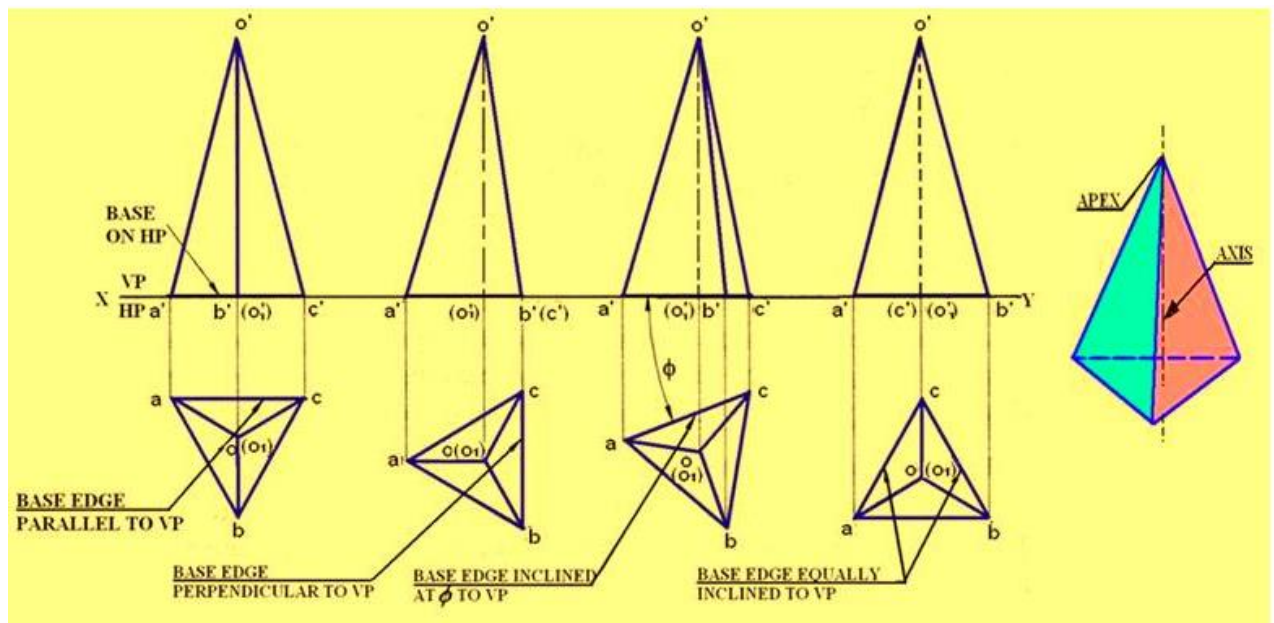


Figure 12. Projections of a triangular Pyramid resting on its base on HP with different positions.

Projections of a solid with the axis perpendicular to VP

When a solid is placed with its axis perpendicular to VP, the base of the solid will always be perpendicular to HP and parallel to VP. Hence in the front view, base will be projected in true shape. Therefore, when the axis of the solid is perpendicular to VP, the front view is drawn first and then the top and side views are drawn from it. When a cylinder rests on HP with its axis perpendicular to VP, one of its generators will be on

HP. Figure 13 shows the Front view and Top view of a cylinder and cone resting on HP with their axes perpendicular to VP. In this case one of the points on the circumference of the base will be on XY.

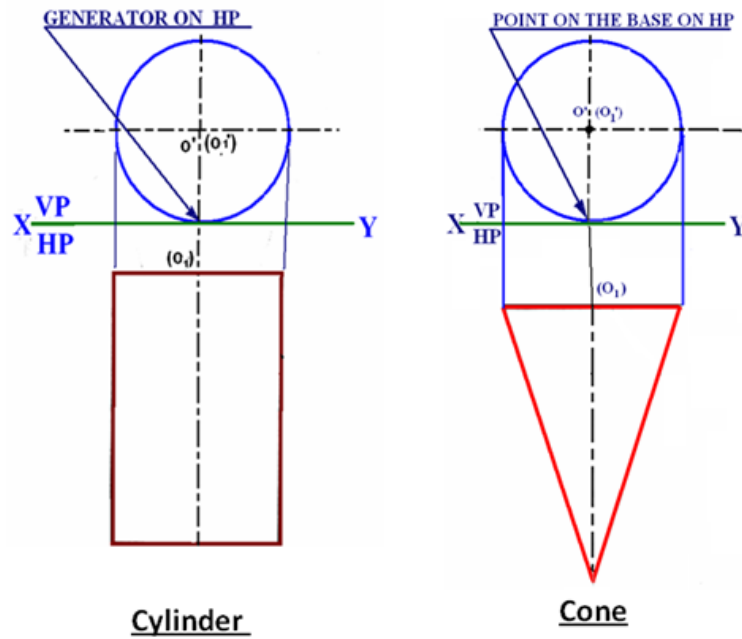


Figure 13 shows the Front view and Top view of a cylinder and cone

Prism may be placed with their axis perpendicular to VP in three different positions. The different positions are shown in figure 14.

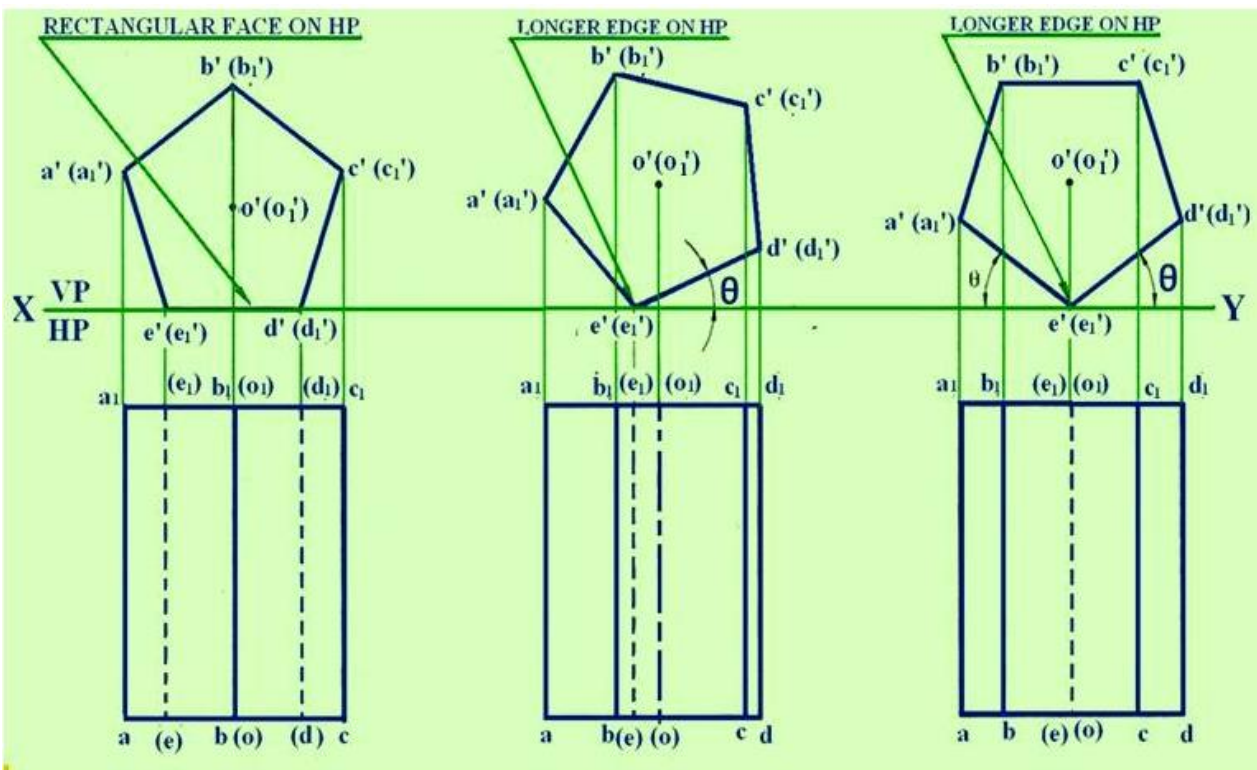


Figure 14. Projections of a pentagonal prism resting on HP and axis perpendicular to VP with different positions.

As shown in Figure 15, pyramid may be placed with their axis perpendicular to VP in three different positions.

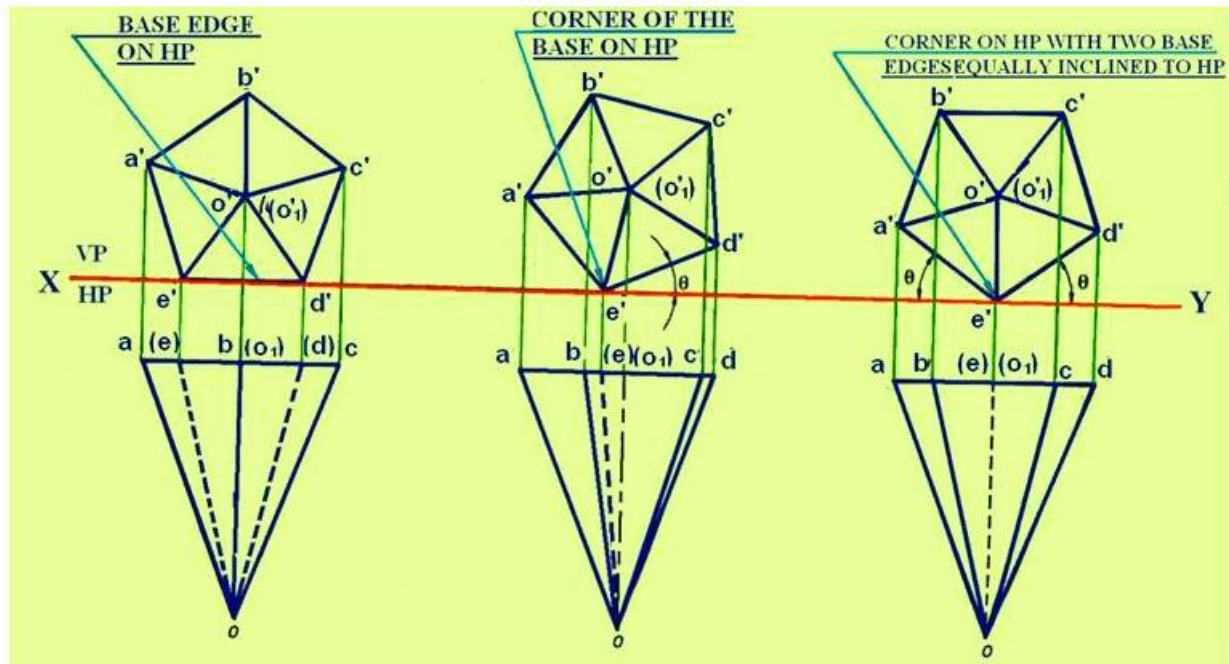


Figure 15. Projections of a pentagonal pyramid resting on HP and axis perpendicular to VP with different positions.

Axis of the solid inclined to HP and parallel to VP

1. When a solid is placed on HP with its axis inclined to HP, the elemental portion of the solid that lies on HP depends upon the type of the solid.
2. When a prism is placed on HP with its axis inclined to it, then it will lie either on one of its base edges or on one of its corners on HP.
3. When a pyramid is placed on HP with its axis inclined to HP, then we will have one of its base edges on HP or one of its base corners on HP or one of its slant edges on HP or one of its triangular faces on HP or an apex on HP.

Methods of drawing the projections of solids

these are two methods for drawing the projections of solids:

1. Change of position method.
2. Auxiliary plane method (Change of reference-line method)

Change of position method

In this method, the solids are placed first in the simple position and then tilted successively in two or three stages to obtain the final position. The following are some of the examples.

a. Method of obtaining the top and the front views of the pyramid when it lies on HP on one of its base edges with its axis or the base inclined to HP.

If the solid is required to be placed with an edge of the base on HP, then initially the solid has to be placed with its base on HP such that an edge of the base is perpendicular to VP, i.e., to XY line in top view preferably to lie on the right side.

When a pentagonal prism has to be placed with an edge of base on HP such that the base or axis is inclined to HP, then initially, the prism is placed with its base on HP with an edge of the base perpendicular to VP and the lying on the right side. In this position, the first set of top and front views are drawn with the base edges (c1)(d1) perpendicular to XY line in the top view. In the front view, this edge c1'(d1') appears as a point. Since the prism has to lie with an edge of the base on HP, the front view of the prism is tilted on the edge c1'(d1') such that the axis is inclined at θ to HP.

Redraw the first front view in the tilted position. Whenever the inclination of axis θ with HP is given, first the base is drawn at $(90^\circ - \theta)$ in the front view, otherwise improper selection of the position of the axis may result in the base edge c1'(d1') lying above or below the XY line. The second top view is projected by drawing the vertical projectors from the corners of the second front view and the horizontal projectors from the first top view. Figure 1 shows the sequence in obtaining the projection of the solid for the above case.

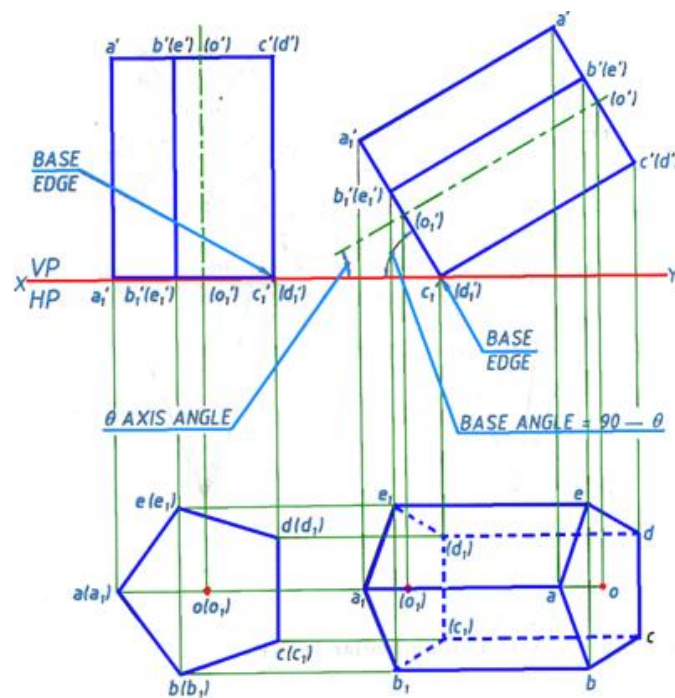


Figure 1. Illustrating the sequence for obtaining the projections of a pentagonal prism placed with an edge of base on HP such that the base or axis is inclined to HP

b. Corner of the base on HP with two base edges containing the corner on which it rests make equal inclinations with HP

When a solid lies on one of its corners of the base on HP, then the two edges of the base containing the corner on which it lies make either equal inclinations or different inclination with HP. Initially the solid should be placed with its base on HP such that an imaginary line connecting the center of the base and one of its corners is parallel to VP, i.e. to XY line in the top view, and preferably to lie on the right side. For example, when a hexagonal prism has to be placed with a corner of the base on HP such that the base or the axis is inclined to HP, then initially the prism is placed with its base on HP such that an imaginary line connecting the center of the base and a corner is parallel to VP and it lies on the right side. In this position, the first set of top and front views are drawn, as shown in step-1 of figure 2. The line (o1)(d1) is parallel to the XY line in the top view.

Since the prism has to lie on one of its corners of the base on HP, the front view of the prism is tilted on the corner d1' such that the axis is inclined at θ to HP. Redraw the front view in the tilted position as shown in Step-2 of figure-2. The base edge is drawn at $(90^\circ - \theta)$ in the front view. The second top view is projected by

drawing the vertical projectors from the corners of the second front view and horizontal projectors from the first top view.

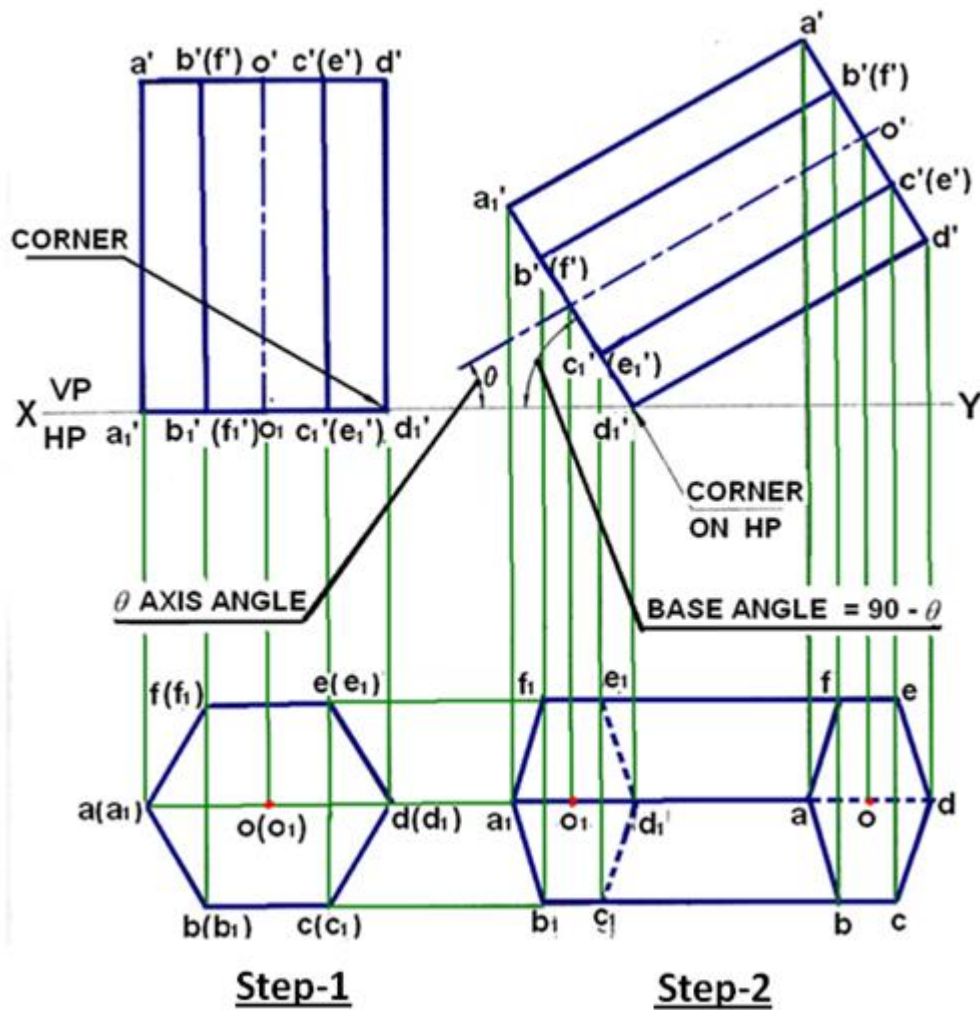


Figure 2: Projections of a prism with a corner of the base on HP and the axis is inclined to HP.

Following the above procedure, the top and front views of the pyramid when it rests on HP on one of its base corners such that the two base edges containing the corner on which it rests make equal inclinations with HP is shown in figure 3.

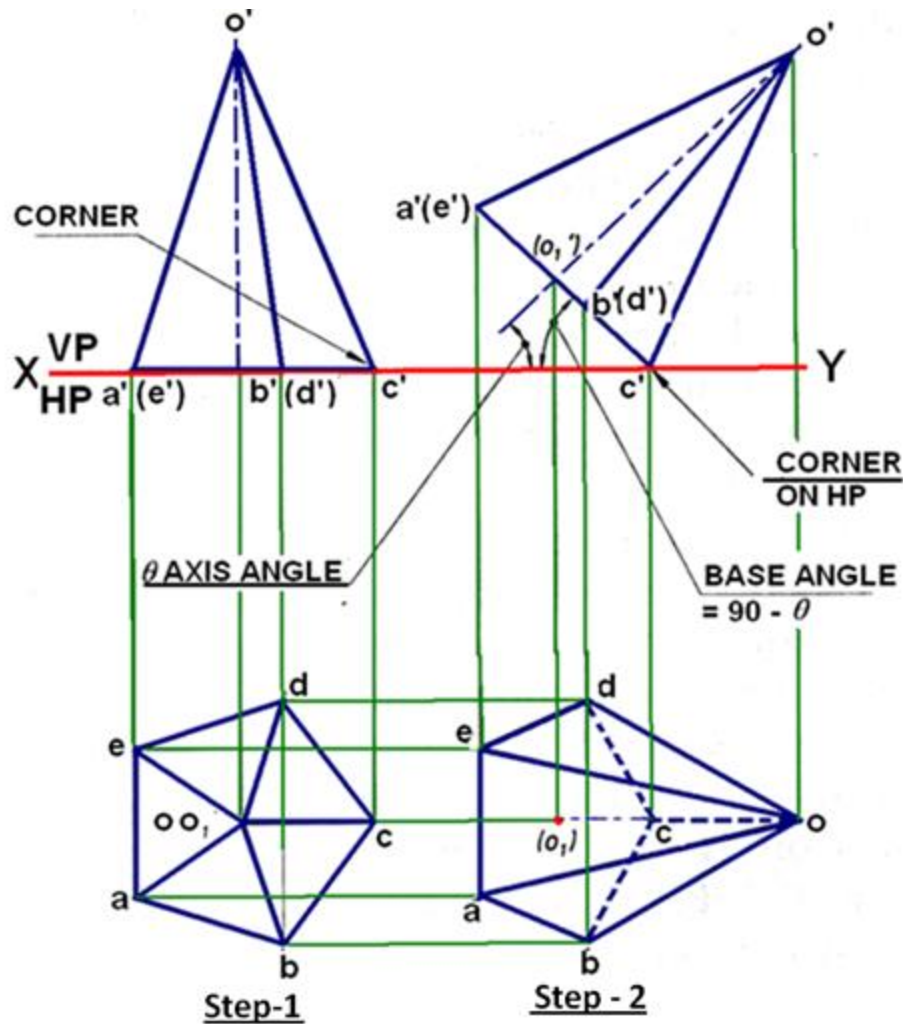


Figure 3 showing the projection of a pyramid resting on HP on one of its base corners with two base edges containing the corner on which it rests make equal inclinations with HP

c. Projections of a pyramid lying on one of its triangular faces on HP

If a pyramid has to be placed on one of its triangular faces on HP, then initially let the pyramid be placed with its base on HP such that the base edge containing that face is perpendicular to VP. (i.e. perpendicular to XY line). Figure 4 illustrates the sequence in obtaining the projections of the pyramid. In the first front view, the inclined line, $o'c'(d')$ represents a triangular face. Redraw the front view such that the line representing the triangular face $o'c'(d')$ lies on HP. Project the top view in this position.

d. Projections of a pyramid lying on one of its slant edge on HP

The sequence of obtaining the projections of a pyramid lying on one of its slant edge on HP is shown in figure 5. In step-1, The FV and TV of the pyramid in the simple projection is drawn such that in the, top view the slant edge (line cd) on which it will rest on ground is drawn parallel to HP (parallel to XY line) in the. In the front view this edge will be line $c'o'$. In step-2, the object is then rotated such that the pyramid lies with its edge $o'c'$ on HP. i.e. in the front view, $o'c'$ lies on XY line. Project the Top view from this Front view.

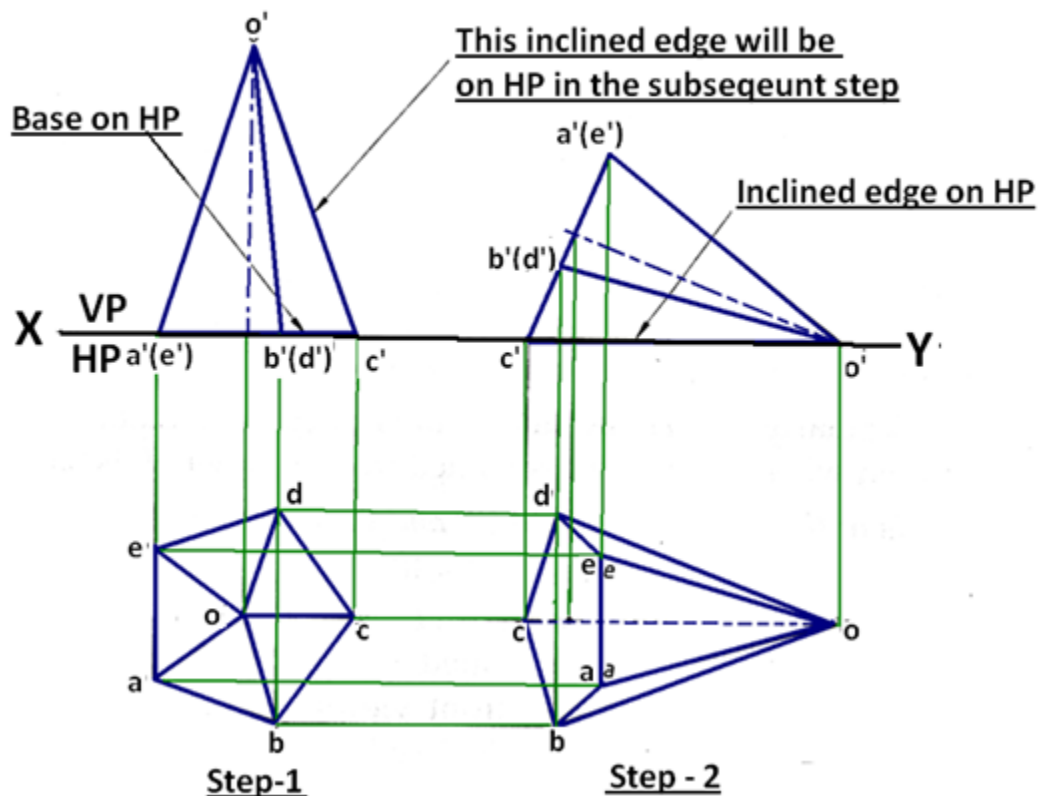


Figure 5. Projections of a pyramid lying on one of its slant edges on HP

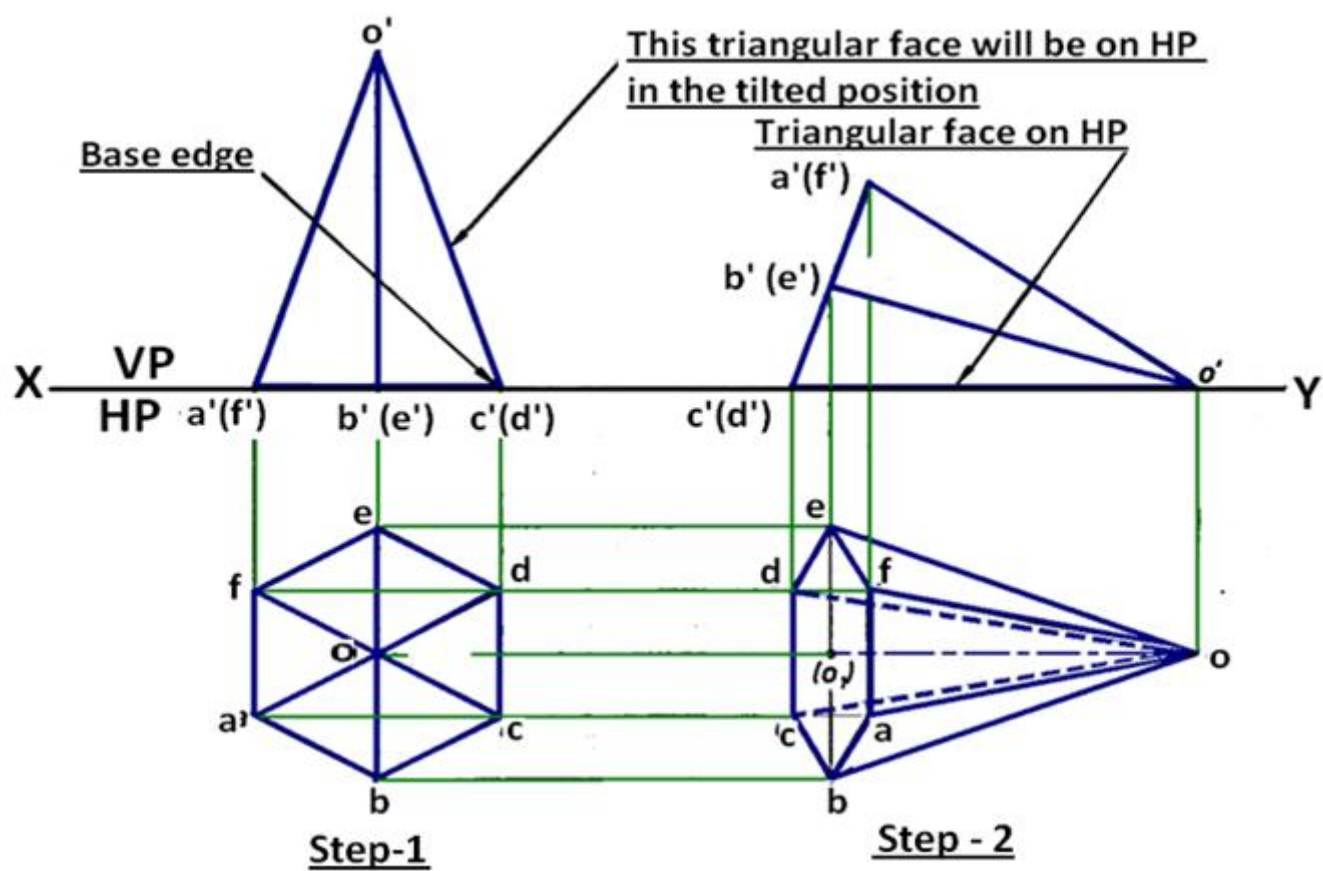


Figure 4 Illustrates the sequence in obtaining the projections a pyramid lying on one of its triangular faces on HP

Problem 1.

A cube of 30 mm sides is held on one of its corners on HP such that the bottom square face containing that corner is inclined at 30° to HP. Two of its adjacent base edges containing the corner on which it rests are equally inclined to VP. Draw the top and front views of the cube.

Solution:

The procedure of obtaining the projections is shown in figure 6.

- The projections of the cube is drawn in the simple position. The cube is assumed to lie with one of its faces completely on HP such that two vertical faces make equal inclinations with VP.
- Draw a square $abcd$ to represent the top view of the cube such that two of its sides make equal inclinations with the XY line, i.e., with VP. Let (a_1) , (b_1) , (c_1) and (d_1) be the four corners of the bottom face of the cube which coincide in the top view with the corners a , b , c and d of the top face.
- Project the front view of the cube. The bottom face $a_1'b_1'c_1'(d_1')$ in the front view coincide with the XY line.
- Now the cube is tilted on the bottom right corner c_1' such that the bottom face $a_1'b_1'c_1'(d_1')$ is inclined at 30° to HP.
- Reproduce the front view with face $a_1'b_1'c_1'(d_1')$ inclined at 30° to the XY line.
- Draw the vertical projectors through all the corners in the reproduced front view and horizontal projectors through the corners of the first top view. These projectors intersect each other to give the corresponding corners in the top view

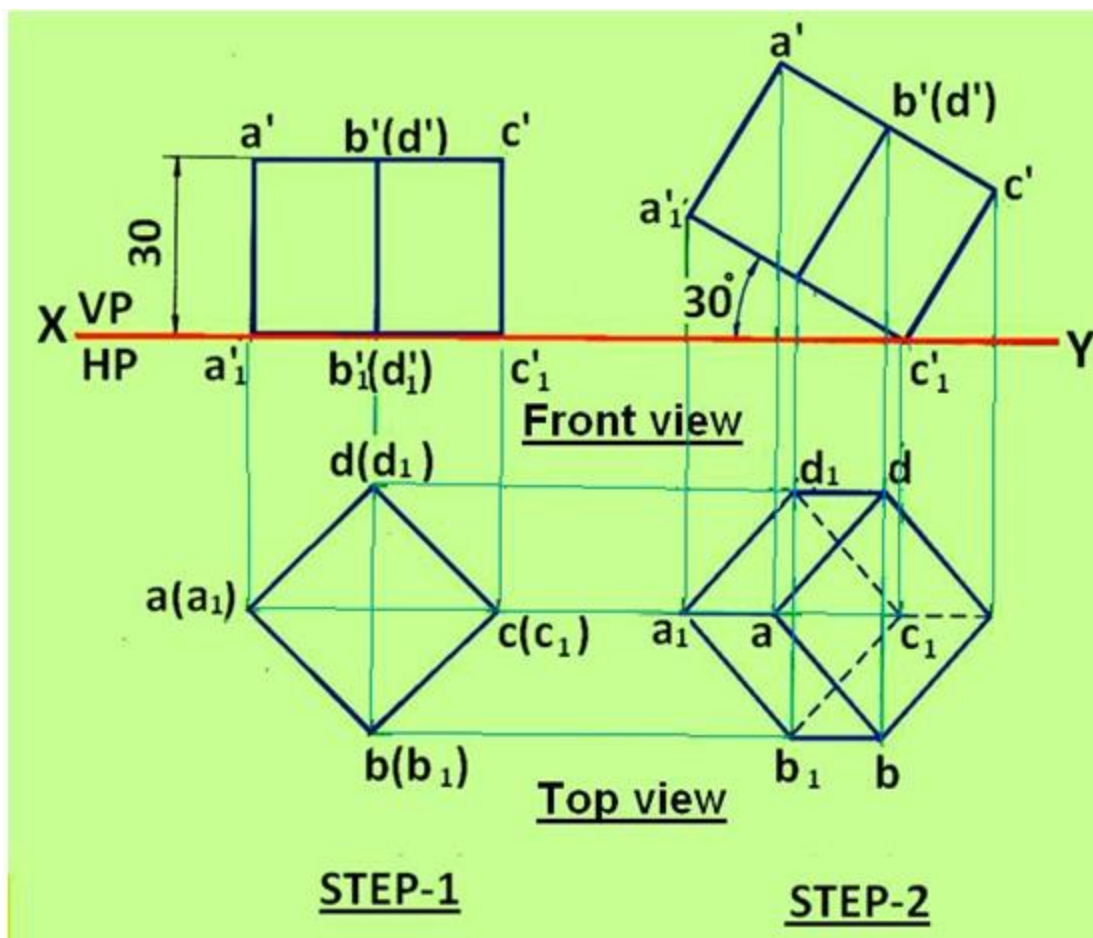


Figure 1. The projections of the cube of problem 1.

Problem-2.

A cube of 30 mm side rests with one of its edges on HP such that one of the square faces containing that edge is inclined at 30° to HP and the edge on which it rests being inclined to 60° to VP. Draw its projections.

Solution.

The procedure of obtaining the projections is shown in figure 7.

- First the TV and FV of the cube is drawn with the cube in the simple position.
- The edge bc is drawn perpendicular to the XY line. In step2, the cube is tilted such that the base of the cube is inclined at 30° to HP.
- The front view is reproduced with $b1' c1' a1' d1'$ inclined at 30° to XY.
- The top view of the cube in step-2 is obtained by drawing projectors mentioned in problem 1.
- In step-3, the top view in step-2 is rotated such that line $c1 b1$ is inclined at 60° to XY line.
- The front view in step-2 is obtained by drawing projectors from the top view in step-3 and Front view in Step-2.

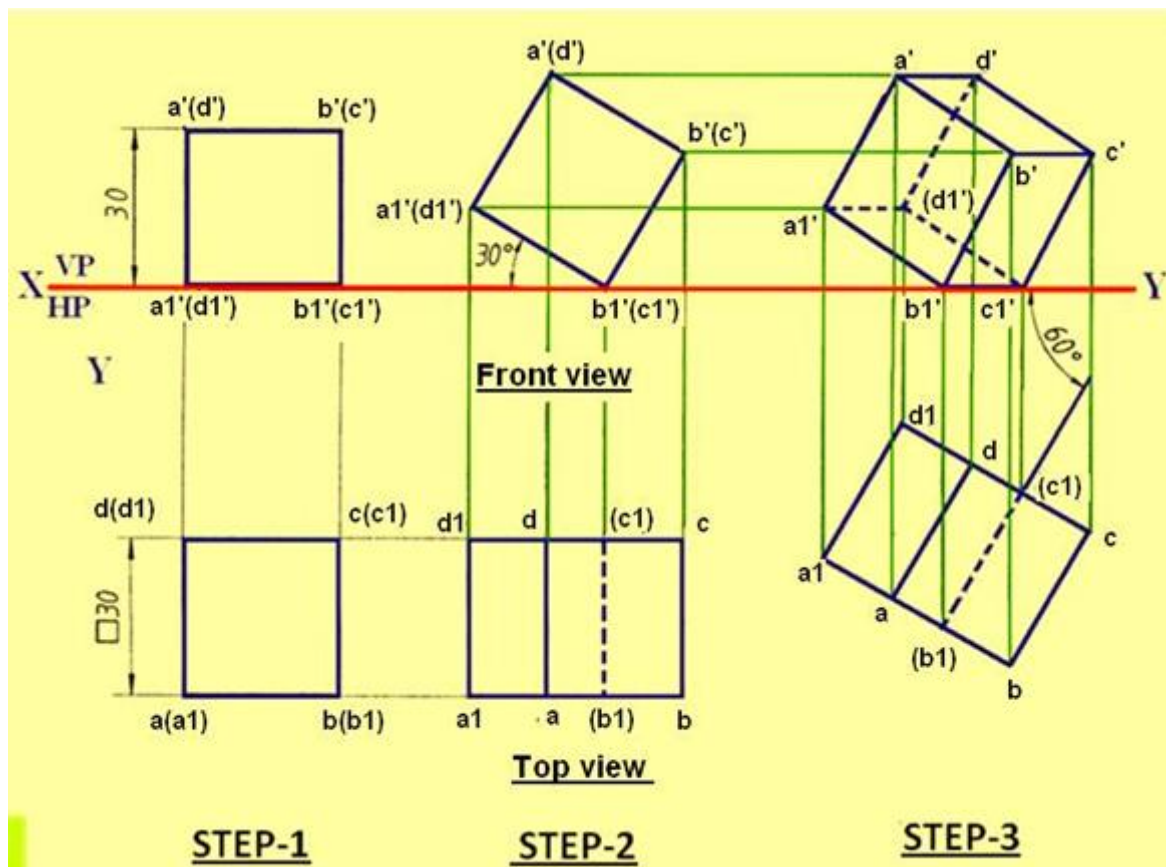


Figure 2. The projections of the cube of problem 2.

Problem 3

An equilateral triangular prism 20 mm side of base and 50 mm long rests with one of its shorter edges on HP such that the rectangular face containing the edge on which the prism rests is inclined at 30° to HP. The edge on which prism rests is inclined at 60° to VP. Draw its projections.

Solution: The procedure of obtaining the projections is shown in figure 8. The prism rests with one of its shorter edges, i.e., triangular or base edge on HP such that the rectangular face containing that edge is inclined at 30° to HP.

- Draw the simple views of the prism when it rests with one of its triangular faces, i.e., base completely lying on HP and also with one of its shorter edges perpendicular to VP, i.e., to XY line.
- The shorter edge $(b_1)(c_1)$ is perpendicular to the XY line. The rectangular face containing the edge $b_1'(c_1')$ is $b_1'b'(c')(c_1')$.
- Now tilt the prism on the edge $b_1'(c_1')$ such that the rectangular face $b_1'b'(c')(c_1')$ is inclined at 30° to the XY line. In this tilted position, project the top view.
- It is seen that the edge b_1c_1 in the top view shown is perpendicular to VP, i.e, to XY line.
- But the edge b_1c_1 has to be inclined at 60° to VP, i.e, to XY line.
- therefore, reproduce the top view with the edge b_1c_1 inclined at 60° to the XY line as shown in the top view.

Project the reproduced top view to get the front view.

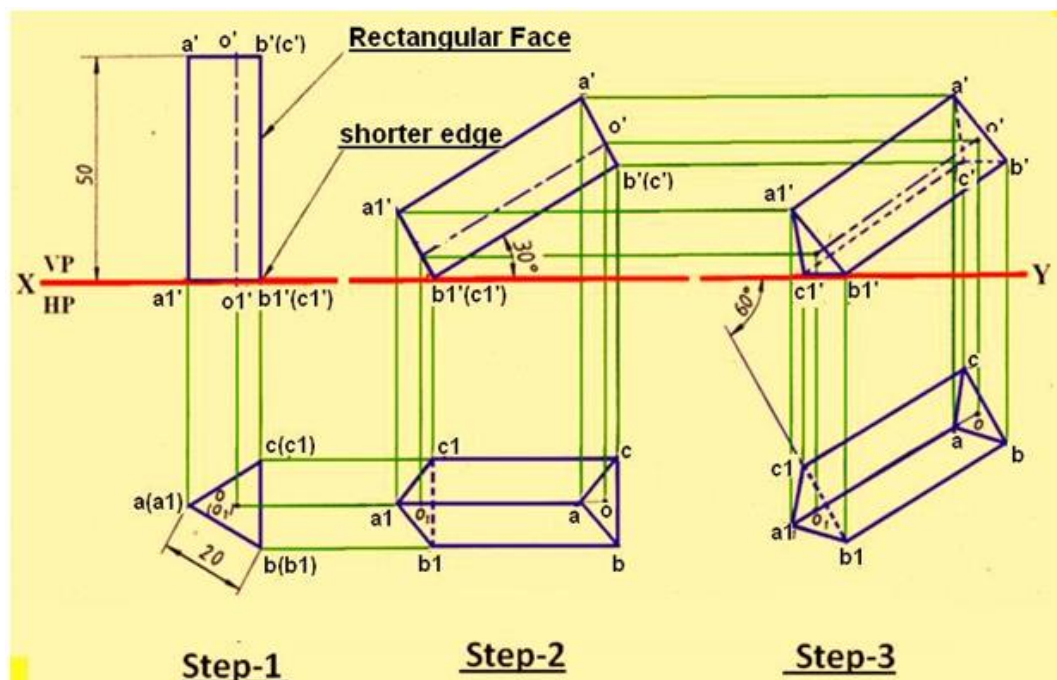


Figure 3. The projections of the triangular prism of problem-3

Problem-4

A hexagonal pyramid has an altitude of 60 mm and side base 30mm. The pyramid rests on one of its side of the base on HP such that the triangular face containing that side is perpendicular to HP. Draw the front and top views.

Solution: The solution the problem is shown in figure 9. In step-1, the pyramid is drawn in the simple position with base edge cd perpendicular to XY line. In Step-2, the Front view is tilted about cd such that line $o'c'd'$ is made perpendicular to XY line. The top view is obtained by drawing projectors from the top view of step 1 and front view in step-2.

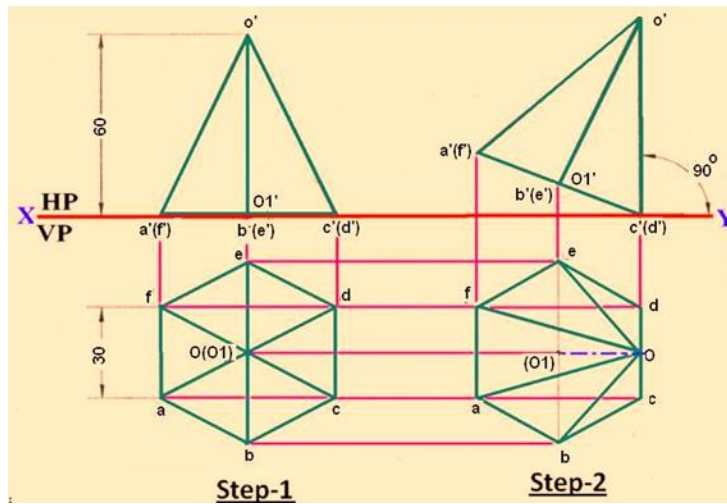


Figure 4. The projections of the hexagonal pyramid of problem-4.

Problem-5

Draw the top and front views of a rectangular pyramid of sides of base 40x 50 mm and height 70 mm when it lies on one of its larger triangular faces on HP. The longer edge of the base of the triangular face lying on HP is inclined at 60° to VP in the top view with the apex of the pyramid being nearer to VP.

Solution:

The solution the problem is shown in figure 5. The projectors are obtained in 3-steps as illustrated in the figure.

- In the first step, the solid is projected in the simple position with base BC perpendicular to VP.
- In the second step, the solid is tilted about the edge BC such that the face BCO is made to lie on the ground.
- The front view is rotated and the top view is projected from the front view and the top view in the first step.
- In step-3, the top view is rotated such that edge BC is inclined at 60° to XY line.
- The Front view is projected using this top view and Front view of Step-2.

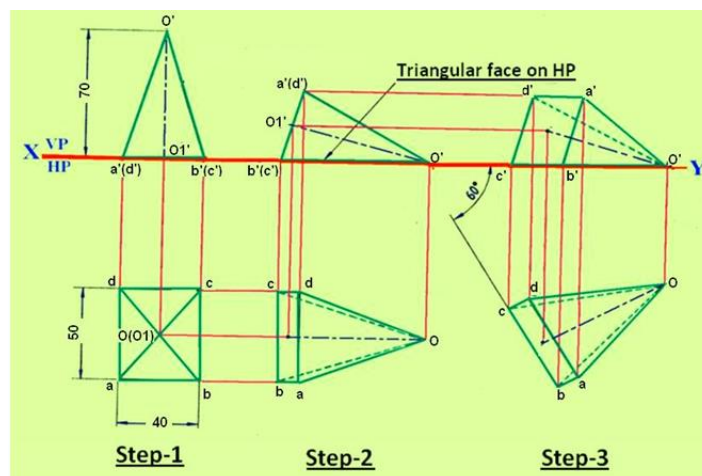


Figure 5. The projections of the rectangular pyramid of problem-5.

Problem-6

A cone of base 80 mm diameter and height 100 mm lies with one of its generators on HP and the axis appears to be inclined to VP at an angle of 40° in the top view. Draw its top and front views.

Solution:

Figure 6 illustrates the procedure for obtaining the projections of the cone. Three steps are involved.

- In step-1, the Top view and Front View of the cone is drawn in the simple position.
- The base circle is divided into 12 equal parts.
- These points are joined with the apex to obtain the respective generators.
- In step-2, the cone is tilted such that the cone lies on one of its generators in the HP. i.e. the generator $g'o'$ is made to coincide with the XY line.
- The top view of the object in this condition is drawn by drawing projectors.
- In step-3, the cone is tilted such that in the top view the axis is inclined at 40° to the XY line.
- The front view of the object is obtained by projection technique.

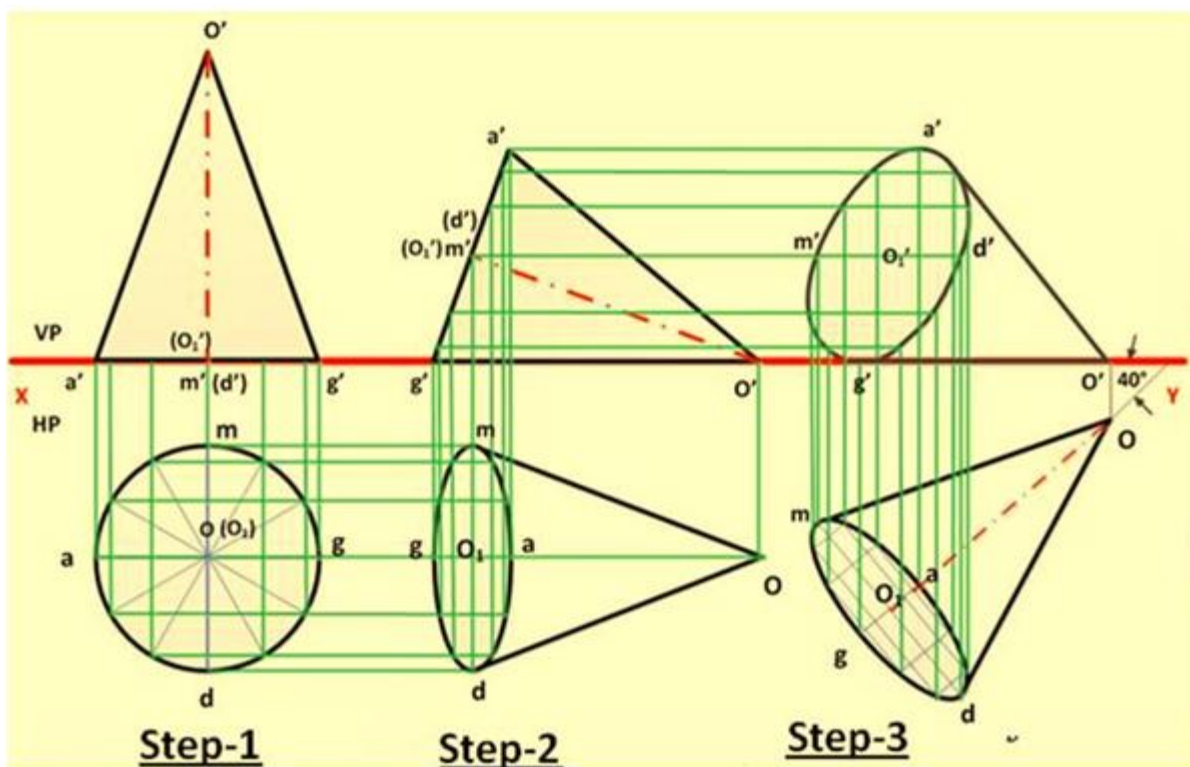


Figure 6. The projections of the cone of problem-6.

Problem-7

Draw the top and the front views of a right circular cylinder of base 45 mm diameter and 60 mm long when it lies on HP such that its axis is inclined at 35° to HP and the axis appears to be perpendicular to VP in the top view.

Solution:

The solution to the problem is illustrated in figure-7. Three steps are involved as shown in the figure.

- In Step-1, the cylinder is drawn in the simple position (resting on the base on HP).

- The circle in the top view is divided in to 12 equal parts and then projected in to the front view.
- In step-2, The Front view is rotated about g_1 such that the axis is inclined at 35° to HP (or XY line).
- The top view is projected from this front view with the help of Top view in step-1. .
- In step 3, the top view is rotated such that axis is perpendicular to XY line.
- The front view is then projected from the top view.

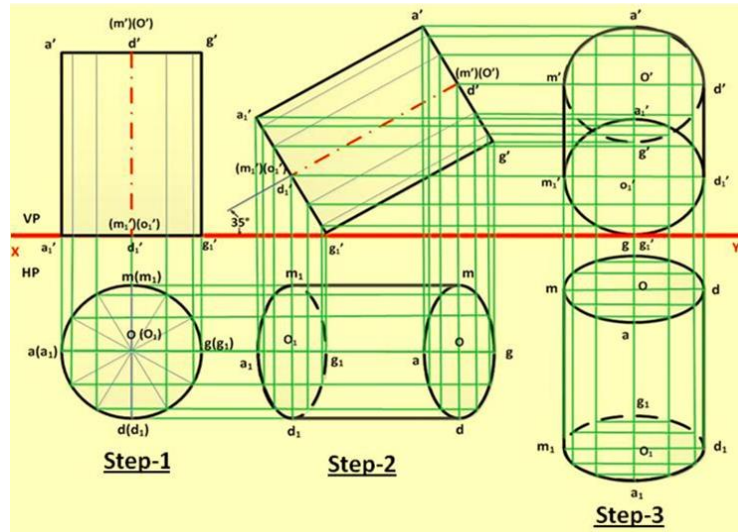


Figure 7. The projections of the cylinder as per problem-7.

Section Views

Introduction

In engineering industries, when the internal structure of an object is complicated, it is very difficult to visualize the object from its orthographic views since there will be several hidden lines. In such case, the internal details are shown by sectional views. Sectional views are an important aspect of design and documentation since it is used to improve clarity and reveal interior features of parts.

Sectional drawings are multi-view technical drawings that contain special views of a part or parts, which reveal interior features. A primary reason for creating a section view is the elimination of hidden lines, so that a drawing can be more easily understood or visualized. Traditional section views are based on the use of an imaginary cutting plane that cuts through the object to reveal interior features. This imaginary cutting plane is controlled by the designer and are generally represented by any of the following:

- Full section view, where the section plane go completely through the object. Example shown in figure 1.
- Half section view, where the section plane go half-way through the object. Example shown in figure 2.
- Offset section, where the sectional plane bent through the features that are not aligned. Example shown in figure 3.
- Broken-out section where the section go through part of the object. Example shown in figure 4.

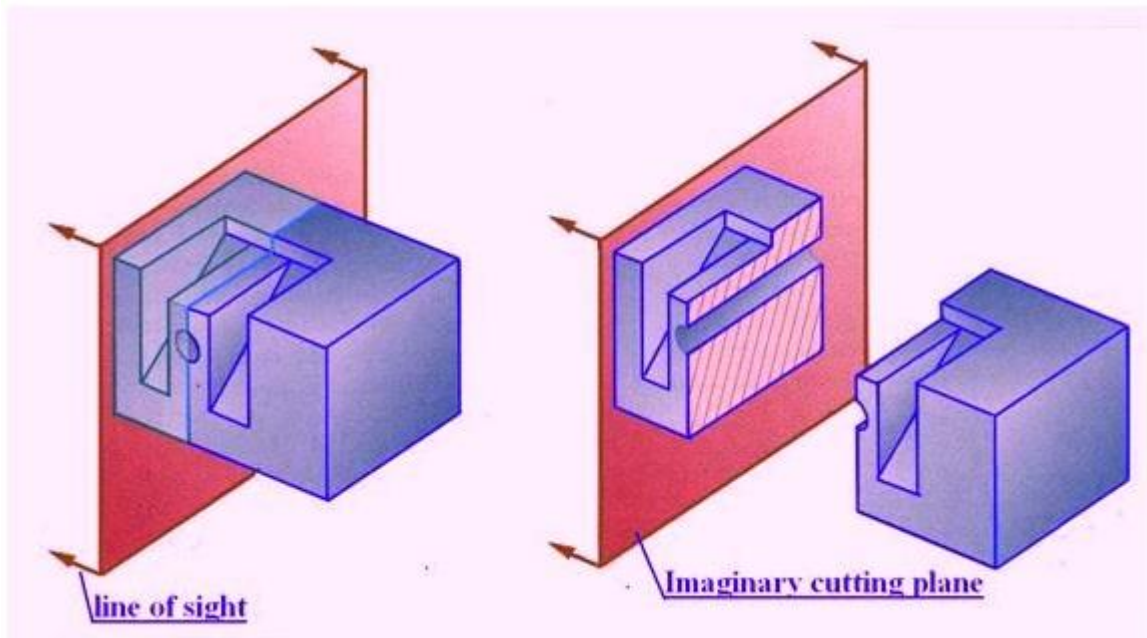


Figure 1. Illustrates a full Section view

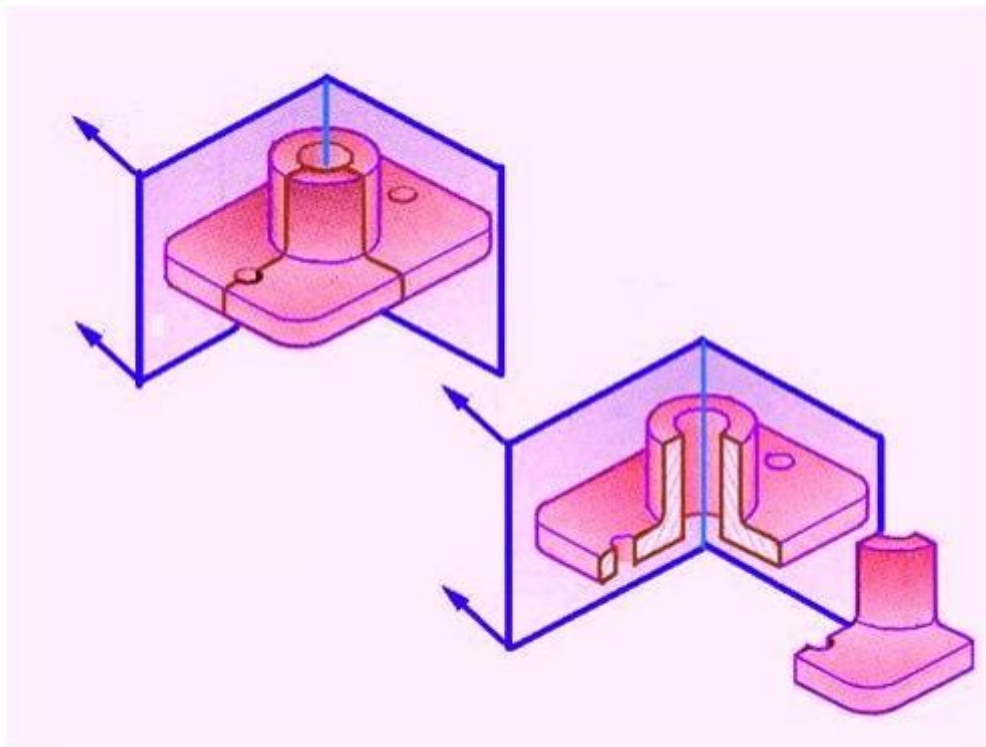


Figure 2. Illustrating a half section view

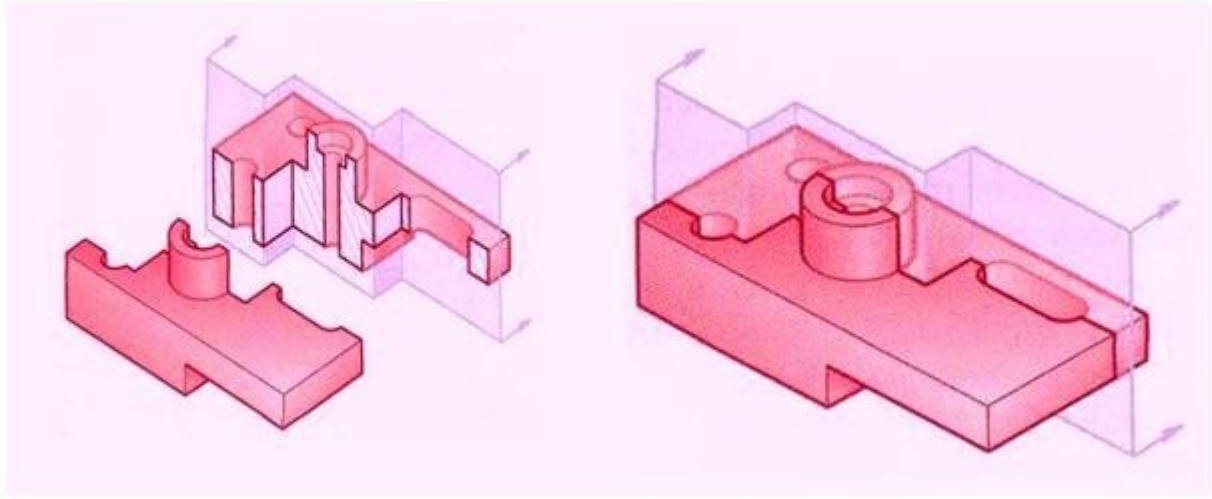


Figure 3. Illustrating an offset section

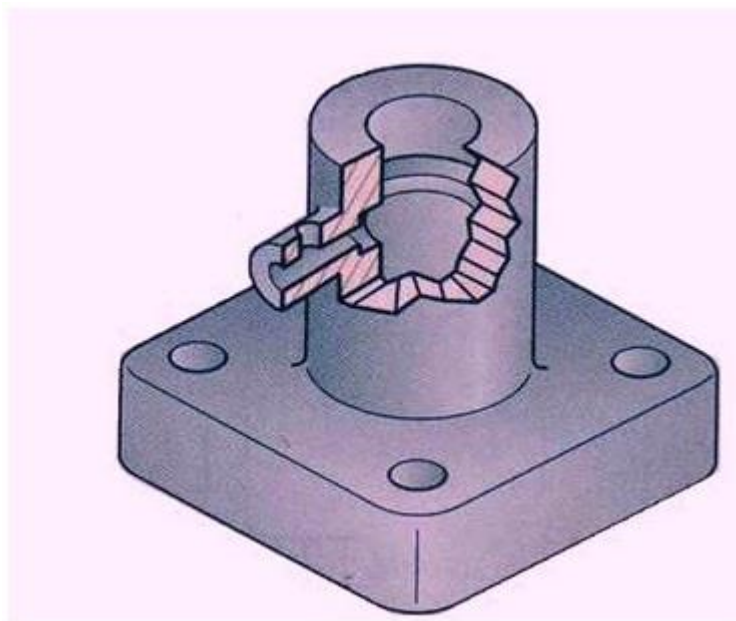


Figure 4. Illustrating a Broken-out Section

An important reason for using section views is to reduce the number of hidden lines in a drawing. A section view reveals hidden features without the use of hidden lines. Adding hidden lines to a section view complicates the drawing defeating the purpose of clarifying with a section. As illustrated in figure 5, there are times, when a minimum number of hidden lines are needed to represent features other than the primary one shown by the section. In the figure, the through and through cavity may not be represented properly since it could be mistaken with a circular hole.

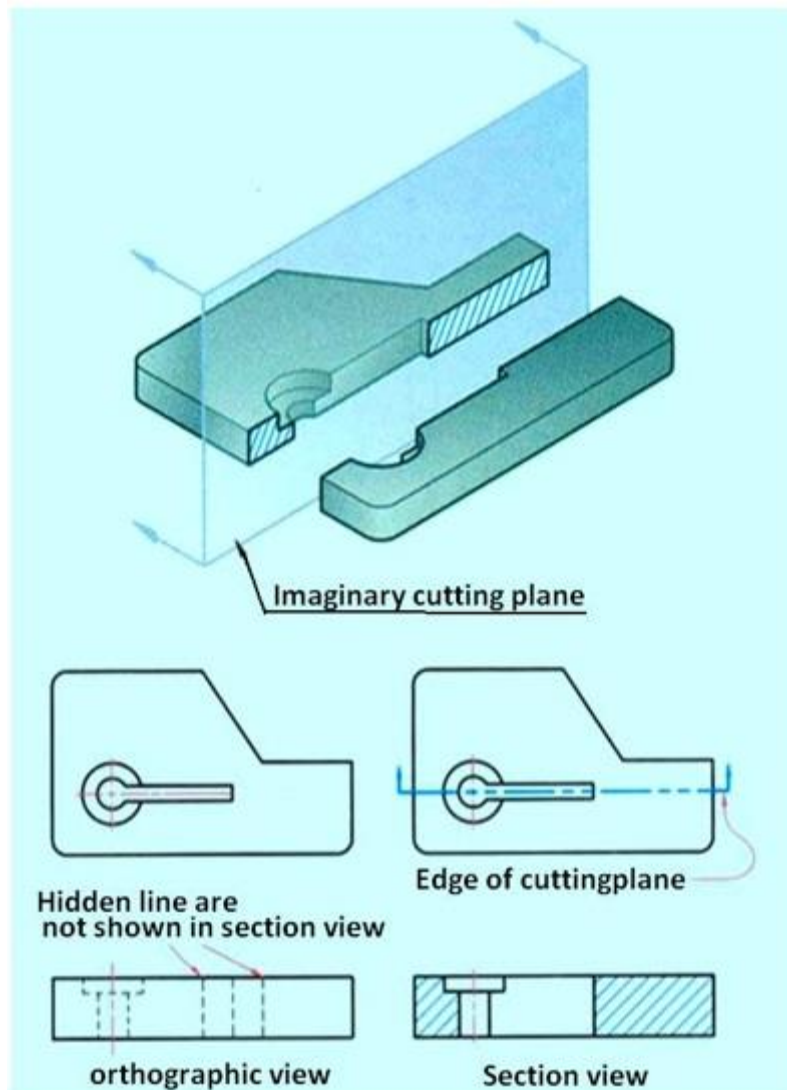


Figure 5 Illustration of the need of section view.

Figure 6 illustrates a full section view of an object. As shown in the correct drawing, the section lined areas are bounded by visible lines, never by hidden lines, because the bounding lines are visible in the section view. Visible surfaces and edges that represent a change of planes behind the cutting plane are drawn as lines in a section view.

Figure 6 shows a section view for which the cutting plane passes through the center of a counter bored hole. A line represents the change of planes between the drilled and counter bored holes and is an example of a visible feature that is behind the cutting plane.

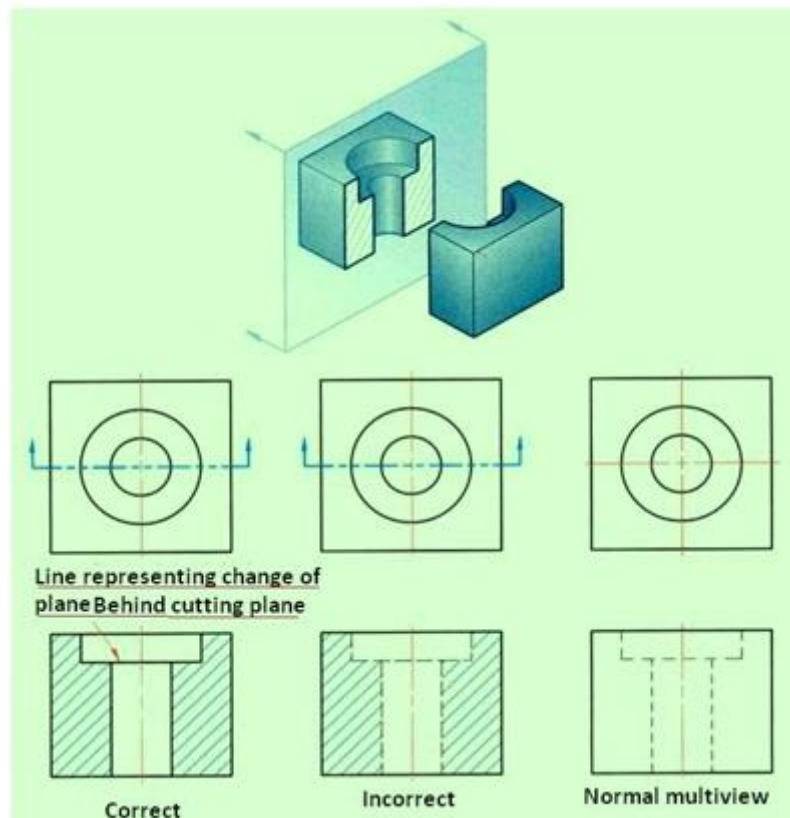


Figure 6 Shows a correct and incorrect representation along with the multi view drawing

Visualization of Section Views

Figure 7 is multi-view drawing of a part that may be difficult to visualize in its 3-D form, because of the many hidden lines.

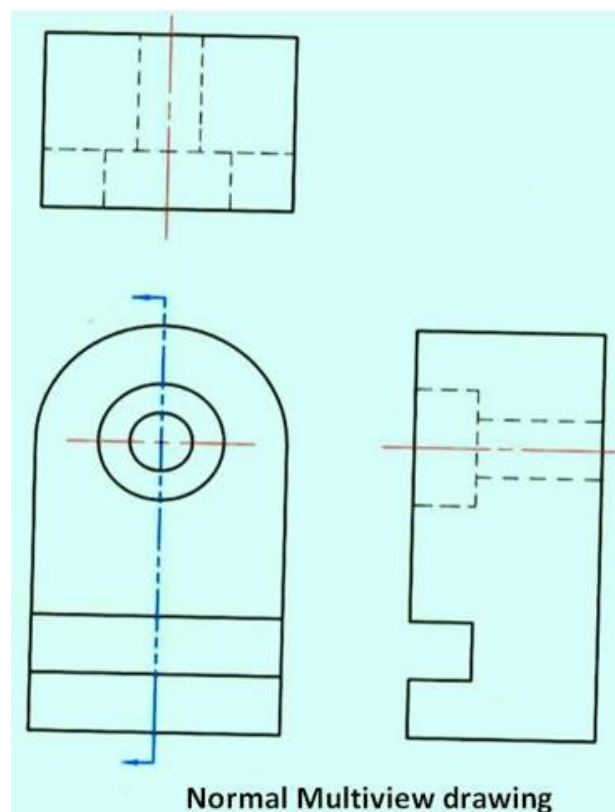


Figure 7 A multi view drawing of an object

A section view is created by passing an imaginary cutting plane vertically through the center of the part. Figure 8 is a 3D representation of the part after it is sectioned. The section view more clearly shows the interior features of the part. In the left corner of the figure, the cutting plane arrows, in the front view, point to the left, to represent the direction of sight for producing a right side view in full section. The direction of the arrow can also be thought of as pointing toward the half of the object being kept. The right half of the object is "removed" to reveal the interior features of the part. The line of sight for the section view is perpendicular to the cut surfaces, which means they are drawn true size and shape in the section view. Also, no hidden lines are drawn and all visible surfaces and edges behind the cutting plane are drawn as object lines. The corners of the section view are numbered as shown in the right hand figure so that they can be compared with the orthographic section view.

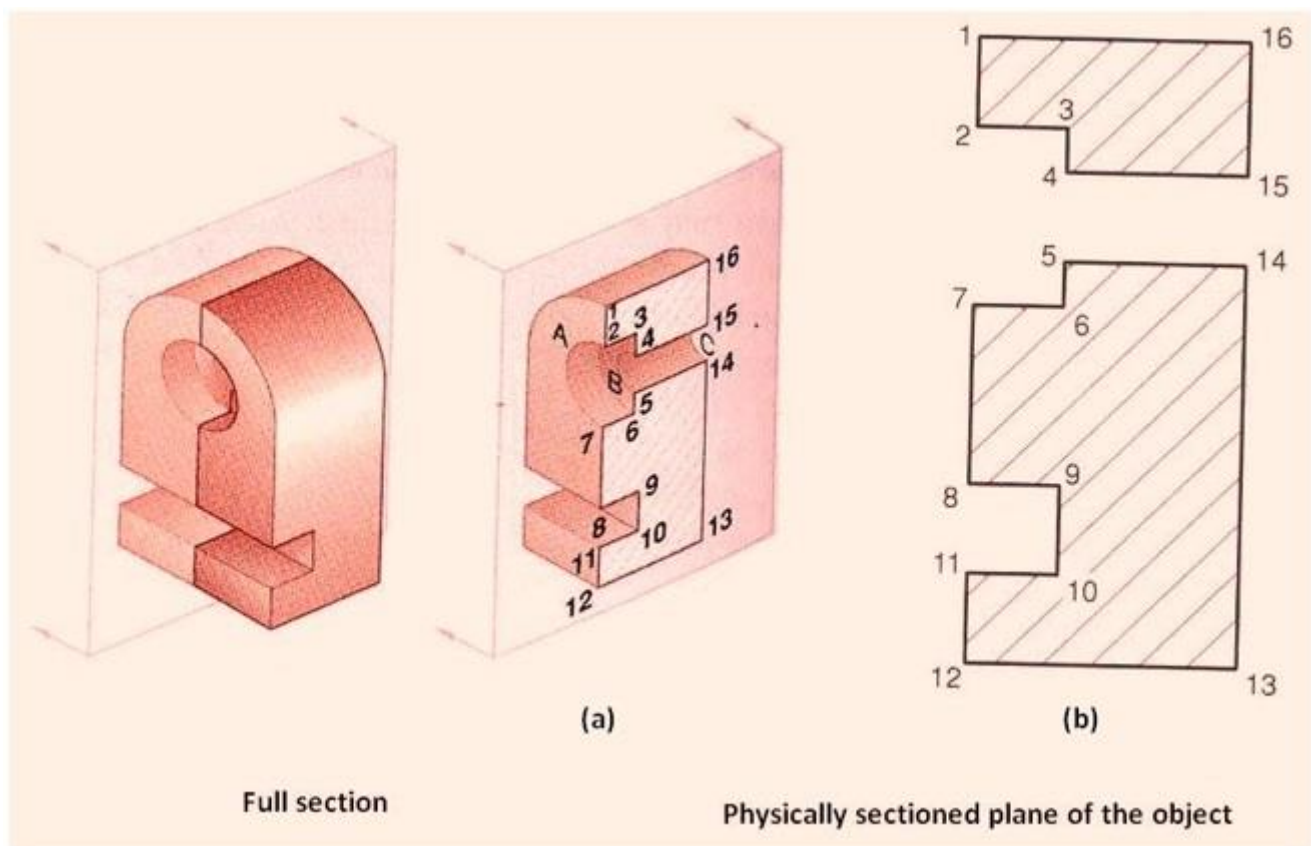


Figure 9 showing a full section and the physically sectioned plane of the object

The representation of the section view of the object shown in figure 9 is shown as (b) in figure 10. The section view in figure (a) shows only those surfaces touched by the cutting plane. Since conventional practice requires that features behind the cutting plane be represented, the change of planes between the two holes in the counter bored hole are shown in figure (b). If the section is viewed along the line of sight identified by the arrows in figure (c), arcs A, B, and C will be visible and should be represented as lines. In figure (b), the lines are 2-7,4-5,15-14. The counter bore and through holes are represented as rectangular features 2-7-6-3, and 4-5-14-15. All the surfaces touched by the cutting plane are marked with section lines. Because all the surfaces are the same part, the section lines are identical and are drawn in the same direction. The center line is added to the counter bored hole to complete the section view.

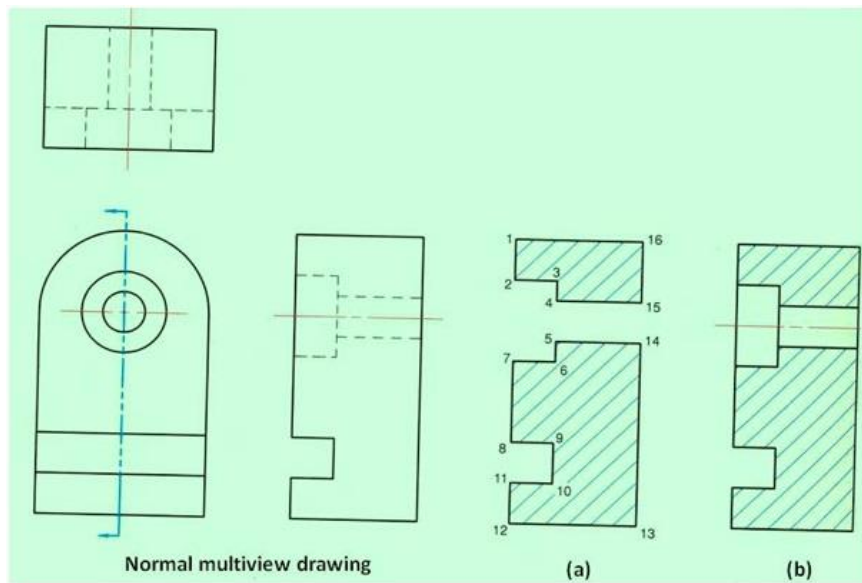


Figure 10. Showing the object, orthographic view, sectioned surface and sectional view.

CUTTING PLANE LINES: The cutting plane line show the exact line along which the cutting plane passes through the object. This represent the edge view of the cutting plane and are drawn in the view(s) adjacent to the section view. This is represented in figure 11. In the figure the cutting plane line is drawn in the top view, which is adjacent to the sectioned front view.

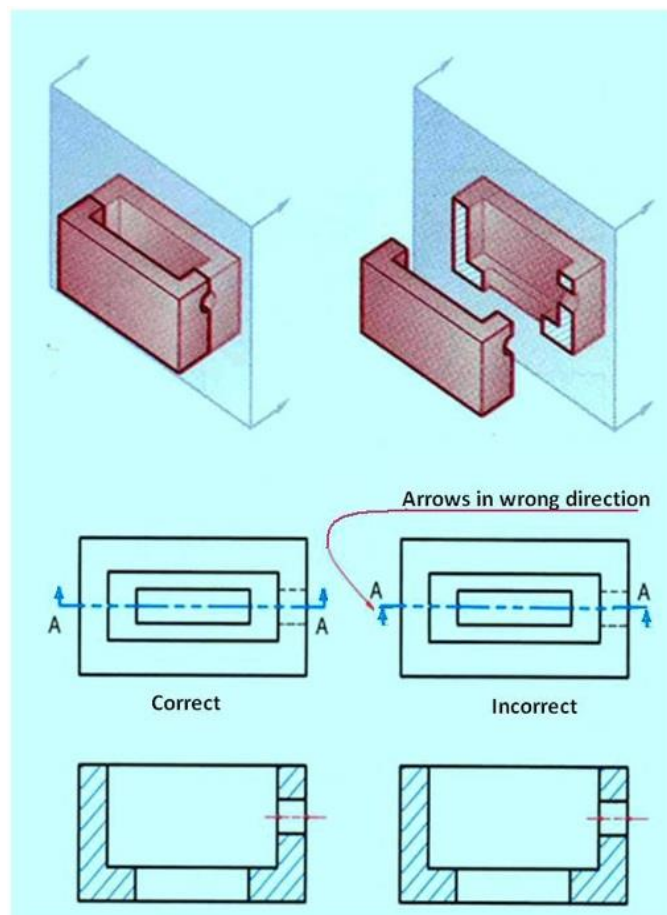


Figure 11 showing the representation of a cutting plane line

Cutting plane lines are *thick* (0.7 mm) dashed lines, that extend past the edge of the object 6 mm and have line segments at each end drawn at 90 degrees and terminated with arrows. The arrows represent the direction of the line of sight for the section view and they point away from the sectioned view. Two types of lines are acceptable for cutting plane lines in multi-view drawings. The normal representation of cutting plane lines are shown in figure 12. Line B-B is composed of alternating long and two short dashes, which is one of the two standard methods. The length of the long dashes varies according to the size of the drawing, and is approximately 20 to 40 mm. For a very large section view drawing, the long dashes are made very long to save drawing time. The short dashes are approximately 3 mm long. The open space between the lines is approximately 1.5 mm. Capital letters are placed at each end of the cutting plane line, for clarity or when more than one cutting plane is used on a drawing. The second method used for cutting plane lines is shown by line C-C, which is composed of equal-length dashed lines. Each dash is approximately 6 mm long, with a 1.5 mm space between. Capital letters are placed at each end of the cutting plane line, for clarity or when more than one cutting plane is used on a drawing. The second method used for cutting plane lines is shown by line C-C, which is composed of equal-length dashed lines. Each dash is approximately 6 mm long, with a 1.5 mm space between.

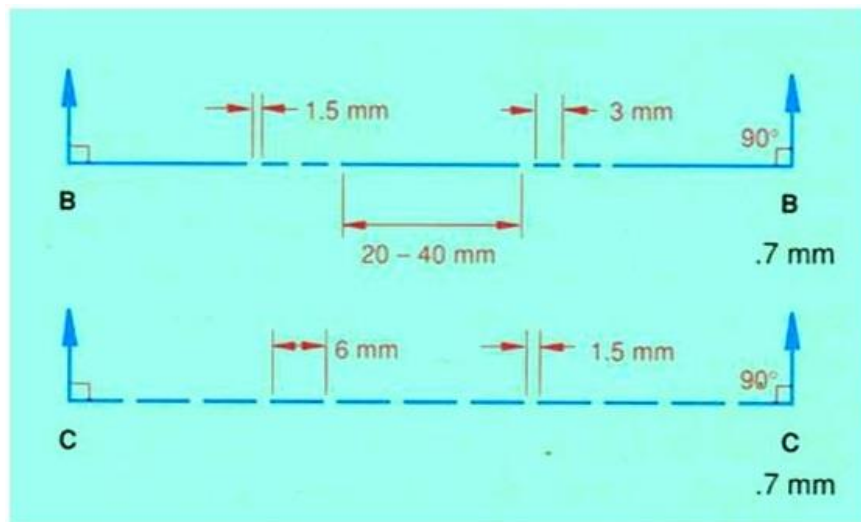


Figure 12 Normal representation of cutting plane lines.

Placement of Cutting Plane Lines

Cutting plane lines are only added to a drawing for clarity. If the position of the cutting plane is obvious, the line need not be drawn. Also, if the cutting plane line is in the same position as a center line, the cutting plane line has precedence.

In figure 10, the cutting plane appears as an edge in the top view and is normal in the front view; therefore, it is a **frontal cutting plane**. The front half of the object is "removed" and the front view is drawn in section. If the cutting plane appears as an edge in the front view and is normal in the top view, it is a horizontal cutting plane. The top half of the object is "removed" and the top view is drawn in section. Figure 13 shows a horizontal cutting plane.

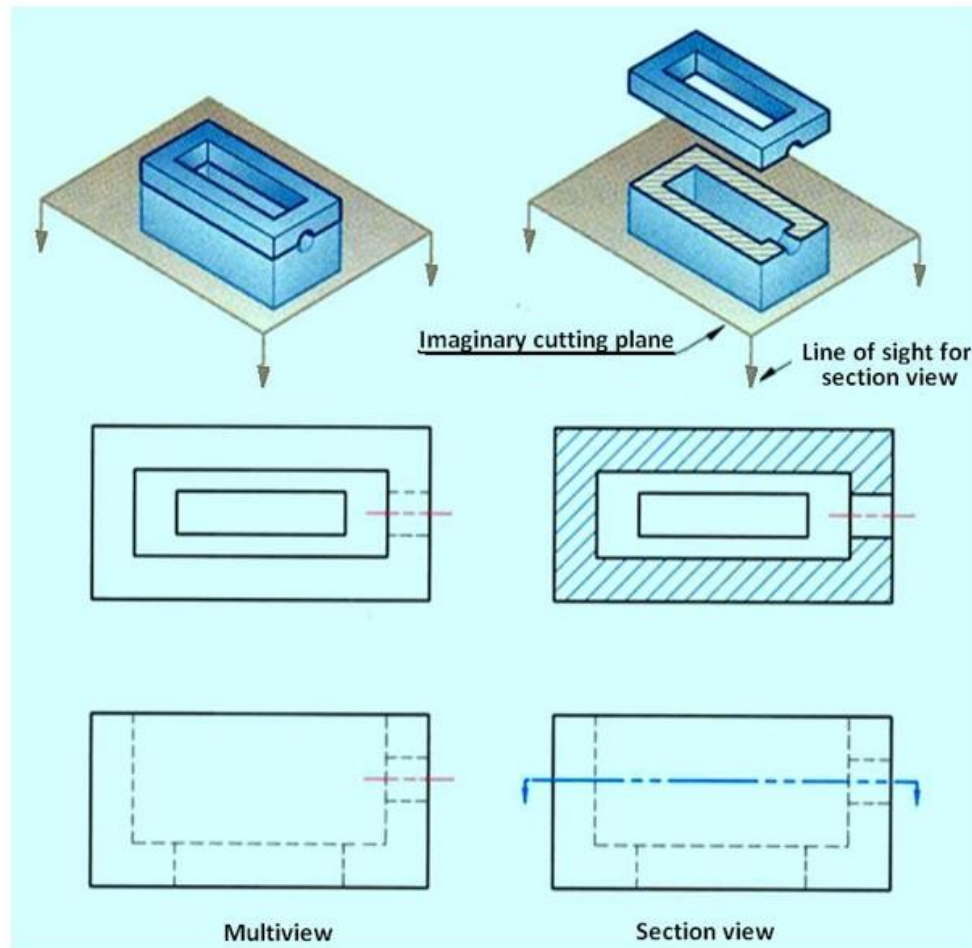


Figure 13. A horizontal cutting plane.

If the cutting plane appears as an edge in the top and front views and is normal in the profile view, it is a profile cutting plane. The left (or right) half of the object is "removed" and the left (or right) side view is drawn in section. A profile cutting plane is shown by BB in figure 14.

Multiple sections can be done on a single object, as shown in the figure 14. In this example, two cutting planes are used: one a horizontal and the other a profile cutting plane. Both cutting planes appear on edge in the front view, and are represented by cutting plane lines A-A and B-B, respectively. Each cutting plane will create a section view, and each section view is drawn as if the other cutting plane did not exist.

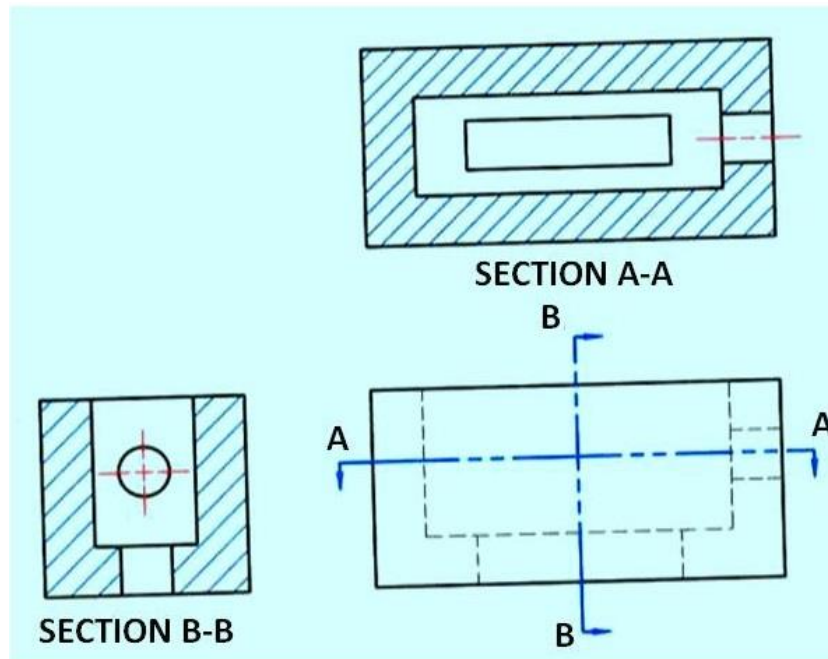


Figure 14. Shows a profile cutting plane.

Section Line Practices

Section lines or cross-hatch lines are added to a section view to indicate the surfaces that are cut by the imaginary cutting plane.

Different section line symbols can be used to represent various types of materials. However, there are so many different materials used in engineering design that the general symbol (i.e., the one used for cast iron) may be used for most purposes on engineering drawings.

The actual type of material required is then noted in the title block or parts list or as a note on the drawing. The angle at which lines are drawn is usually 45° degrees to the horizontal, but this can be changed for adjacent parts shown in the same section. Also the spacing between section lines is uniform on a section view.

Material Symbols

The type of section line used to represent a surface varies according to the type of material. Symbols generally used for various materials are shown in figure 15. However, the general purpose section line symbol used in most section view drawings is that of *cast iron*. The specific type of steel to be used will be indicated in the title block or parts list. Occasionally, with assembly section views, material symbols are used to identify different parts of the assembly.

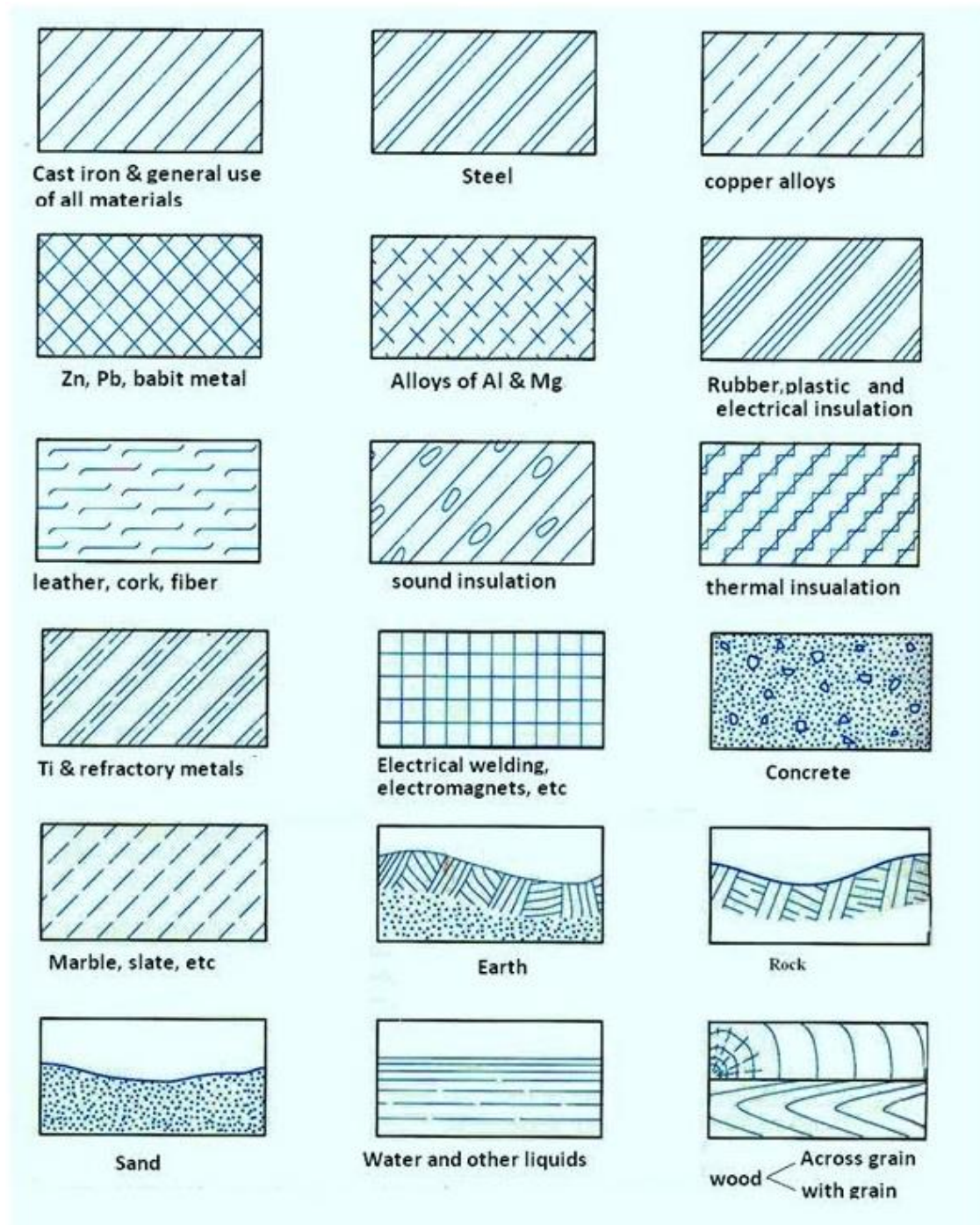


Figure 15. General symbols used to represent various materials in section view.

The general purpose cast iron section line is drawn at a 45° and spaced 1.5 mm to 3 mm or more, depending on the size of the drawing. As a general rule, use 3mm spacing. Section lines are drawn as thin (.35 mm) black lines. The section lines should be evenly spaced and of equal thickness, and should be thinner than visible lines. Also, do not run *section* lines beyond the visible outlines or stop them too short. The correct and incorrect method of drawing section lines are shown in figure 16.

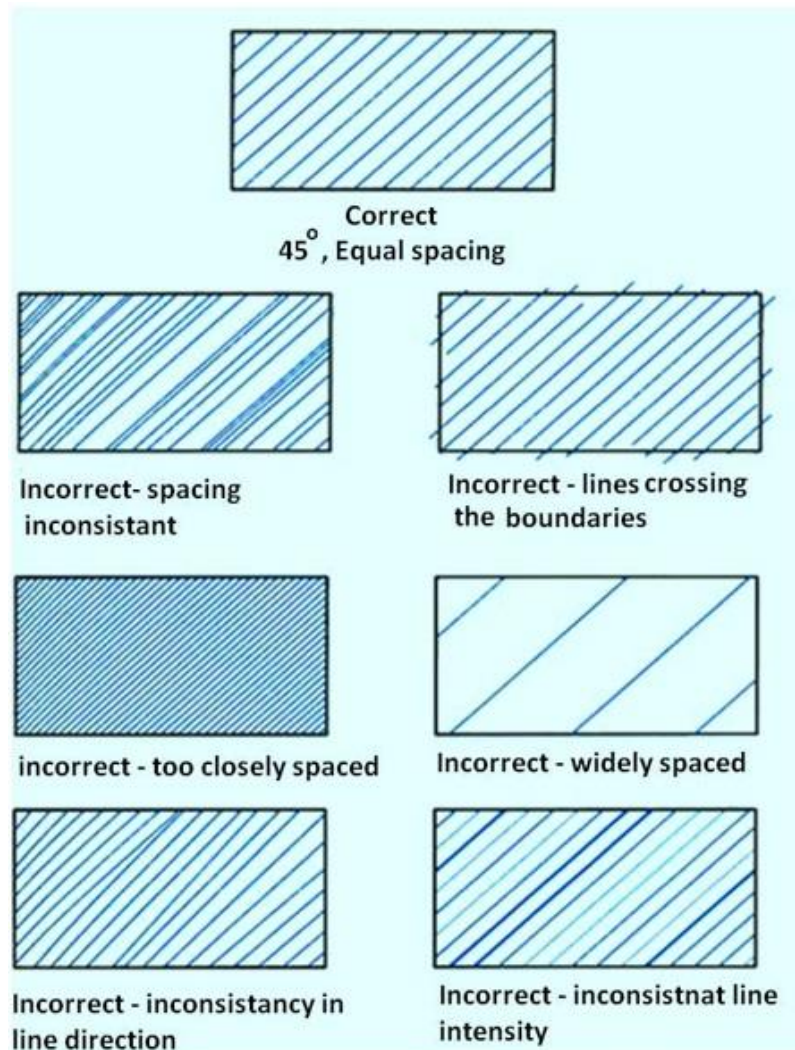


Figure 16. Representation of section line.

Section lines should not run parallel or perpendicular to the visible outline. If the visible outline to be sectioned is drawn at 45°, the section lines are drawn at a different angle, such as 30° as shown in figure 17.

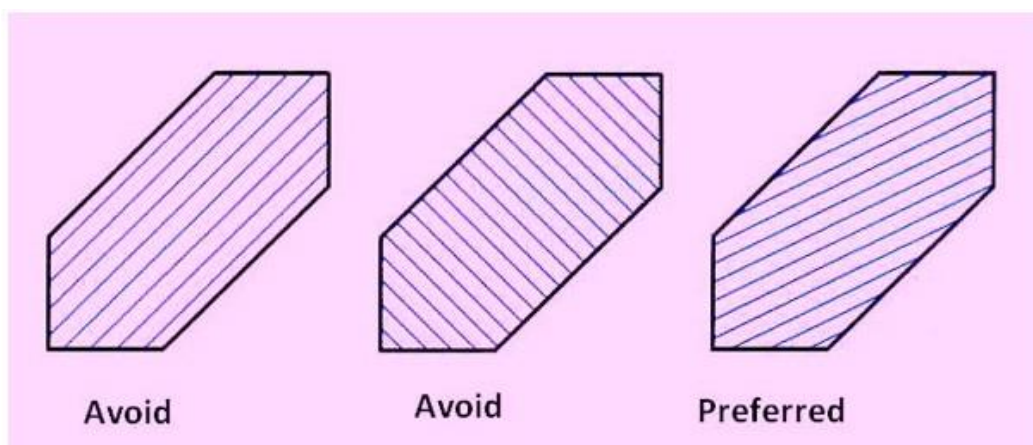


Figure 17 showing the direction of section lines with respect to the visible outline of the object.

Avoid placing dimensions or notes within the section lined areas. If the dimension or note must be placed within the sectioned area, omit the section lines in the area of the note as shown in figure 17.



Figure 18 Placement of dimensions or notes within the section lines.

An outline section view is created by drawing partial section outlines adjacent to all object lines in the section view as shown in figure 19. For large parts, outline sectioning may be used to save time.

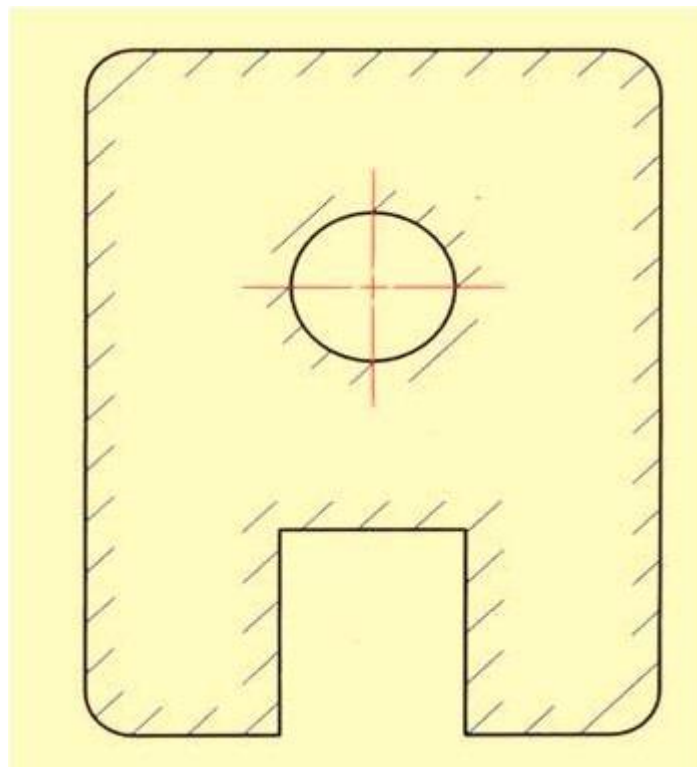


Figure 19. Shows an outline section view.

Very thin parts such as washers and gaskets are not easily represented with section lines, so conventional practice calls for representing the thin part in solid black. Figure 20 shows the section of a gasket drawn as solid black.

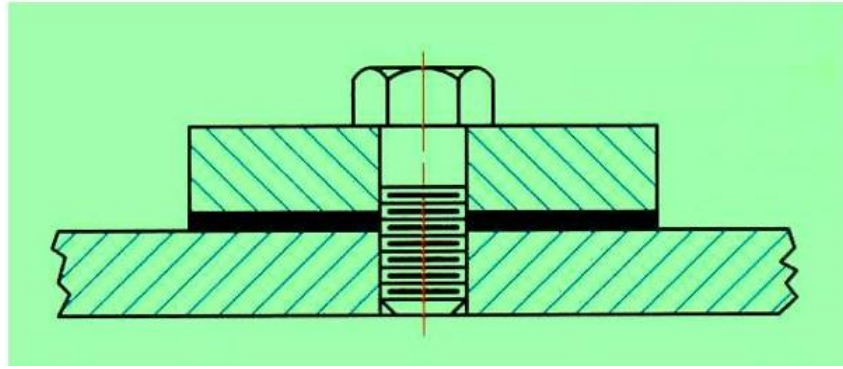


Figure 20. Representation of section view of a very thin part (e.g. gasket)

Full Section View

A full section view is made by passing the imaginary cutting plane completely through the object. As shown in figure 21, all the hidden features intersected by the cutting plane are represented by visible lines in the section view. Surfaces touched by the cutting plane have section lines drawn at a 45-degree angle to the horizontal. Hidden lines are omitted in all section views unless they must be used to provide a clear understanding of the object. The top view of the section drawing shows the cutting plane line, with arrows pointing in the direction of line of sight to view the sectioned half of the object. In a multi-view drawing, a full-sectioned view is placed in the same position that an un-sectioned view would normally occupy, I.e., a front section view would replace the traditional front view.

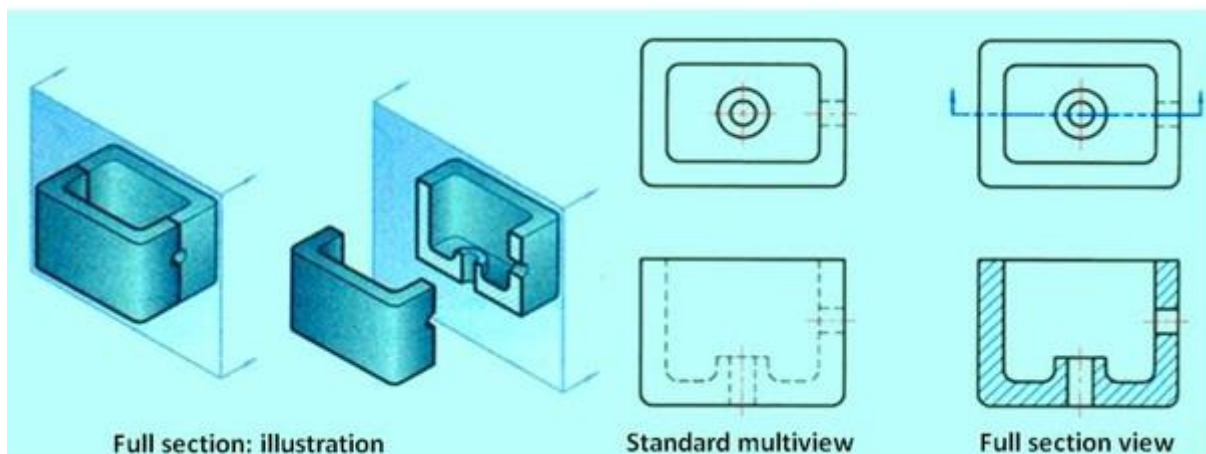


Figure 21 shows a full section view of an object.

Half Section view

Half sections are created by passing an imaginary cutting plane only halfway through an object. Hidden lines are omitted on both halves of the section view. Hidden lines may be added to the un-sectioned half, for dimensioning or for clarity. External features of the part are drawn on the un-sectioned half of the view. A center line, not an object line, is used to separate the sectioned half from the un-sectioned half of the view. The cutting plane line shown in the top view. The cutting plane line in the top view is bent at 90° and one arrow is drawn to represent the line of sight needed to create the front view in section. Half section views are used most often on parts that are symmetrical, such as cylinders. Also, half sections are commonly used in assembly drawings when external features are also to be shown. Figure 22 shows a half section view of an object.

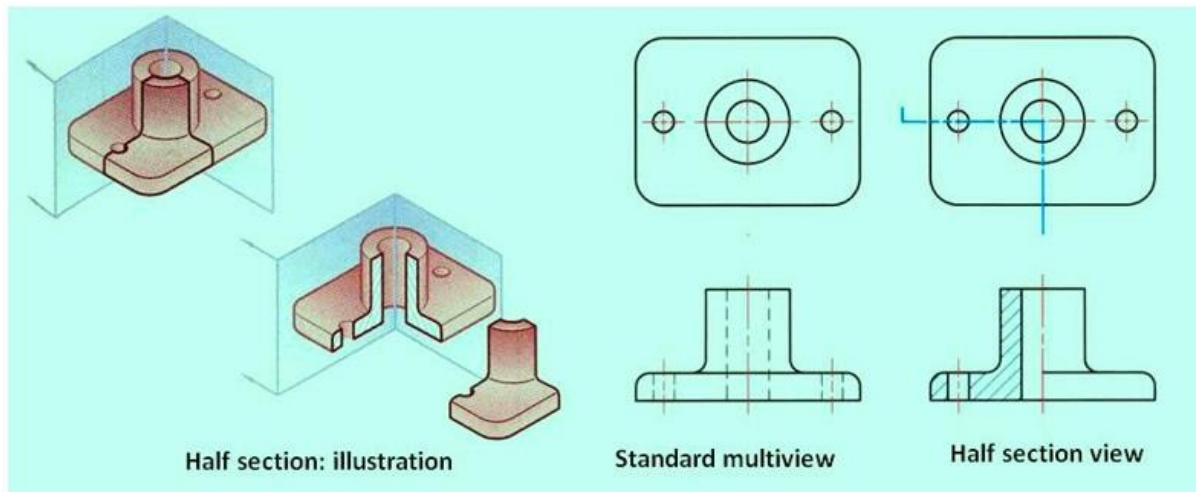


Figure 22 shows the cutting plane passing halfway through an object and one quarter of the object being removed

Broken-Out Sections

A broken-out section is used when only a portion of the object needs to be sectioned. A broken-out section view of an object is shown in figure 23. A break line separates the sectioned portion from the un-sectioned portion of the view. A break line is drawn freehand to represent the jagged edge of the break. No cutting plane line is drawn. Hidden lines may be omitted from the un-sectioned part of the view, unless necessary for clarity. A broken-out section is also used instead of a half or full section view to save time.

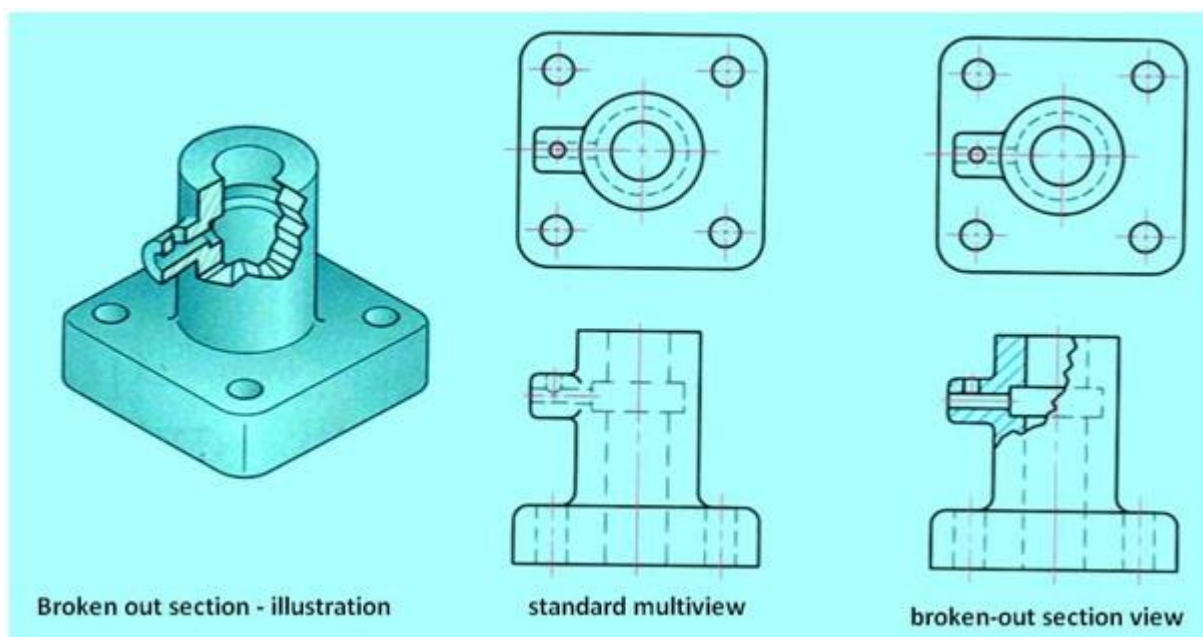


Figure 23 shows the broken out section of an object. Majority of the technical details are also shown.

Section plane parallel to HP

Problem -1: A triangular prism, side of base 30 mm and axis 50 mm long is lying on the HP on one of its rectangular faces with its axis inclined at 30° to the VP. It is cut by a horizontal section plane at a distance of 12 mm above the ground. Draw its front view, side view and sectional top view

Solution:

- The Top view and front view of the triangular prism with respect to the projection planes is drawn first.
- In the Front view, the section plane (line V-T) which is parallel to HP will be a line parallel to and 12 mm above XY line.
- It is evident from figure 7. that the line VT intersects with the edges $a'-b'$ at $1'$, $a'-c'$ at $2'$, $d'-e'$ at $4'$ and $d'-f'$ at $3'$.
- Draw vertical projectors from this points to meet the edges $a-b$ at 1, $a-c$ at 2, $e-d$ at 4 and $d-f$ at 3 respectively.
- When the top portion of the prism is removed, the top view will be obtained as a rectangle, i.e the hatched region bounded by 1-2-3-4.

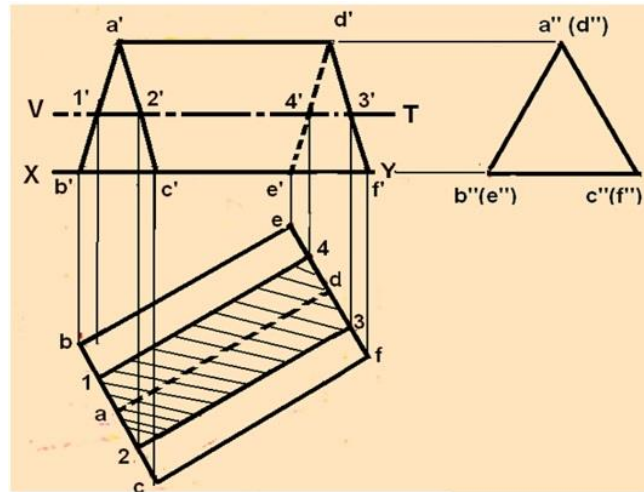


Figure 7. Sectional view of the object in problem-1.

Section plane parallel to HP

Problem-2: A pentagonal pyramid, side of base 30 mm and axis 65 mm long, has its base horizontal and an edge of the base parallel to the VP. A horizontal section plane cuts it at a distance of 25 mm above the base. Draw its front view and sectional top view.

Solution: Figure 8 shows the solution of the problem. The sectional view is obtained in the same manner as in problem-1. Here the intersection of the slant edges of the pyramid with the section plane is projected on to the respective edges in the top view.

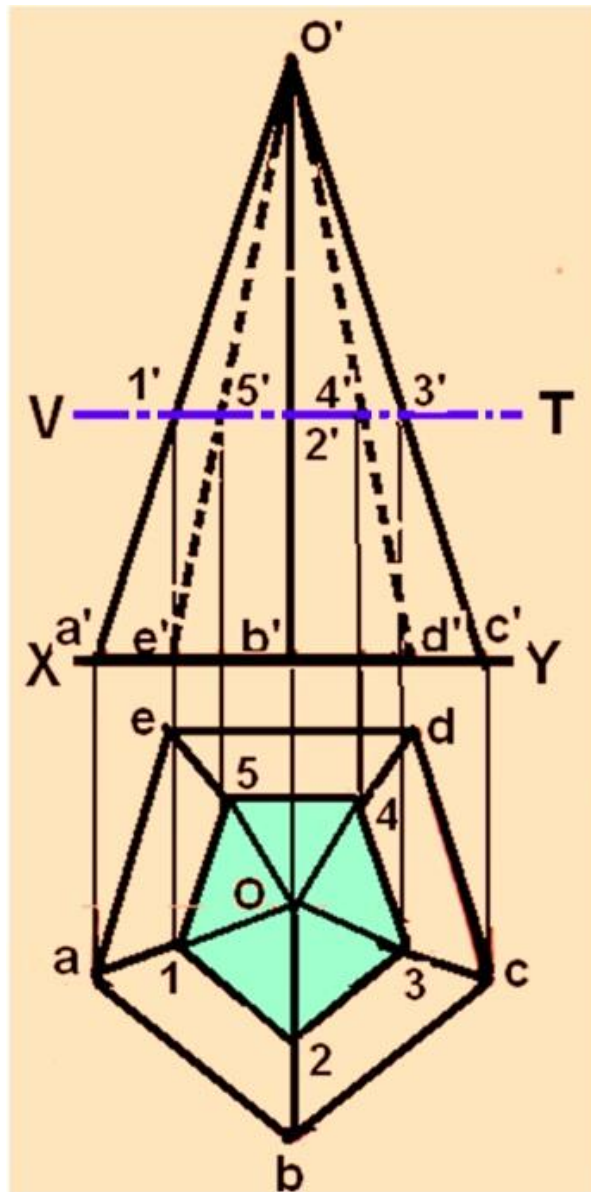


Figure 8. Sectional view for the problem-2.

Section plane Inclined to VP

Problem-3 A pentagonal pyramid has its base on the HP. Base of the pyramid is 30 mm in side, axis 50 mm long. The edge of the base nearer to VP is parallel to it. A vertical section plane, inclined at 45° to the VP, cuts the pyramid at a distance of 6 mm from the axis. Draw the top view, sectional front view and the auxiliary front view on an AVP parallel to the section plane.

Solution:

The solution is shown in figure 9.

- The Front view and top view of the pyramid is first drawn.
- The vertical section which is inclined at 45° to VP is shown by the line HT inclined at 45° to XY line.

- Since the section plane passes through a point 6 mm from the axis of the pyramid, a circle of radius 6 mm is drawn in the top view.
- The line HT is drawn such that it is tangent to this circle.
- The sectional front view of the object is obtained by drawing vertical projectors and its intersection with the respective slant edges in the front view.
- Since point 2 (in the top view) is in the line ob, which is vertical, the projection of point 2' is obtained in the following manner.
- With centre O and radius o-2, draw an arc to intersect the line o-c at 21.
- Project point 21 in the top view to point 2' in the Front view.
- Draw a horizontal projector from 2' to intersect the line o'-b' at 2'.
- The section front view is shown by the hatched area in the front view.
- The auxiliary front view is drawn to obtain the true shape of the section.
- This is obtained by the auxiliary projection method shown in figure 9 .

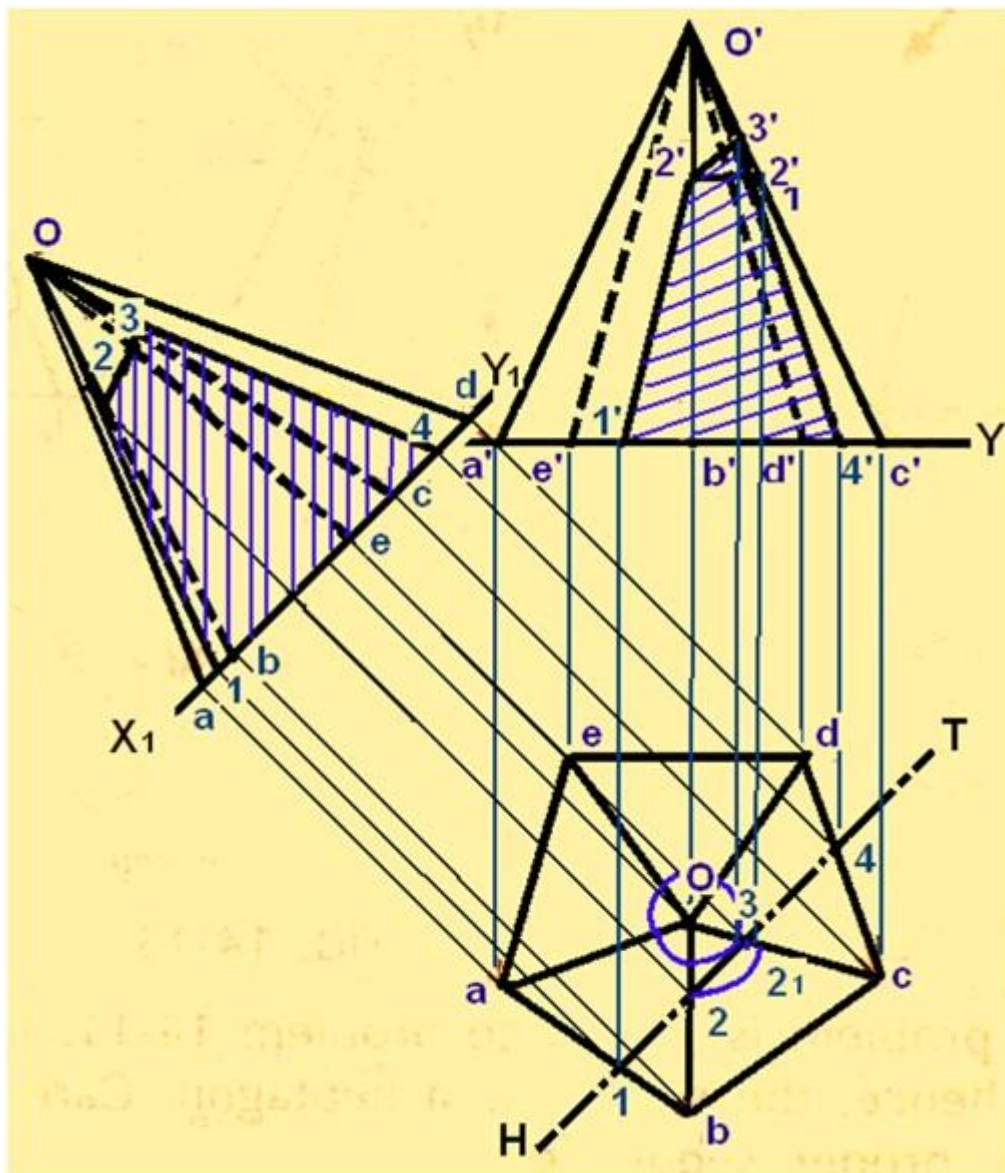


Figure 9. Solution to problem – 3.

Problem:4

A cube of 80 mm long edges has its vertical faces equally inclined to the VP. It is cut by an AIP in such a way that the true shape of the cut part is a regular hexagon. Determine the inclination of the cutting plane with the HP. Draw FV, sectional TV and true shape of the section.

Solution:

The solution is shown in figure 10. Draw TV and FV of the cube. As the true shape of the section is a hexagon, the cutting plane must cut the prism at 6 points. The cutting plane will cut two edges of the top, two edges of the base and two vertical edges. The Points of intersection at two vertical edges will be farthest from each other. These points will represent the two opposite corners of the hexagon and the distance between them will be equal to $b(b_1) - d(d_1)$.

Draw a line $3-6 = b(b_1) - d(d_1)$. Draw a circle with $3-6$ as a diameter. Inscribe a hexagon $1-2-3-4-5-6$ in it as shown. Measure the distance between $1-2$ and $4-5$, i.e., PQ . In the front view, locate $3'$ at the midpoint of $b'(d') - b_1'(d_1')$. With $3'$ as a centre and radius $= \frac{1}{2}(PQ)$, cut arcs on $a'b'$ and $b_1'c_1'$ to locate $1'$ and $4'$ respectively. Join $1'-4'$ for the required cutting plane. Measure θ .

Draw X_1Y_1 parallel to $1'-4'$. Redraw hexagon $1-2-3-4-5-6$ as $1_1-2_1-3_1-4_1-5_1-6_1$ such that pq is parallel to X_1Y_1 . Project all the corners of the hexagon in FV. $2', 6'$ and $5'$ will coincide with $1', 3'$ and $4'$ respectively. Project $1', 2', 3'$, etc., to $1, 2, 3$, etc., on the corresponding edges in TV to obtain the section. 3 and 6 will coincide with $d(d_1)$ and $b(b_1)$ respectively

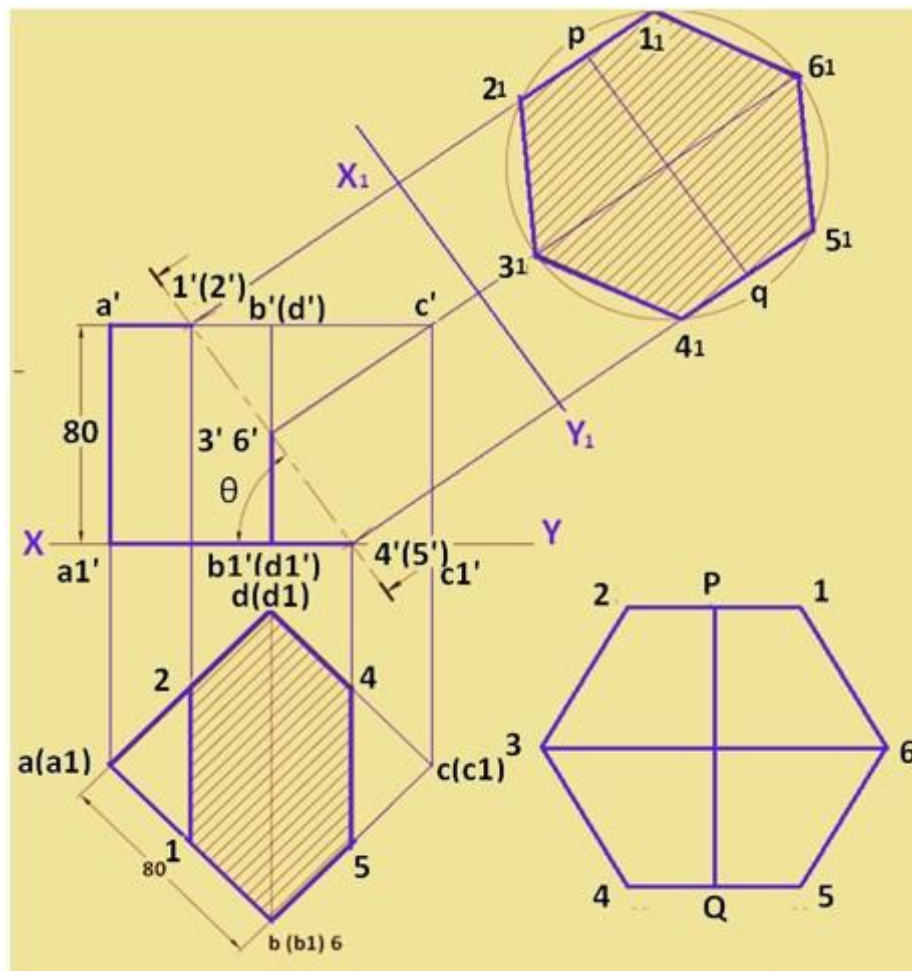


Figure 10. Solution to problem - 4